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Master's Thesis

On constructing abelian extensions of CM fields via CM abelian varieties

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under the supervision of
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Introduction

This is the abstract for a masters' thesis titled "On constructing abelian extensions of CM fields via CM abelian varieties" undertaken at Universität Regensburg, under the supervision of Prof. Guido Kings and Mr. Christoph Fronhoefer.

Understanding extensions of number fields is the basic problem of algebraic number theory as we know it. Class Field theory aims to study a special class of such extensions called abelian extensions, which are defined as Galois extensions of the base field with abelian Galois group. CFT gives a correspondence between the "internal information" i.e., idele class group of the number field and the "external information" i.e., the abelian extensions of the number field. But in general, the established theory does not make this correspondence explicit. Explicit CFT aims to make this correspondence explicit. Hilbert, in his list of 23 (at the time) unsolved problems proposed at the ICM in 1900, posed as the 12th problem: To extend the Kronecker–Weber theorem on abelian extensions of the rational numbers to abelian extensions of any base number field.

Indeed, the Kronecker-Weber theorem is the simplest case of explicit CFT. It states that any abelian extension of $\mathbb{Q}$ is a subextension of some cyclotomic field $\mathbb{Q}(\zeta_n)$ for some $n \in \mathbb{N}$. This classifies all abelian extensions of $\mathbb{Q}$, at the same time giving us a way to explicitly construct all of them. In the same spirit, Kronecker was able to show that if the base field is an imaginary quadratic number field, then one can attach to it an elliptic curve with CM (complex multiplication), such that adjoining torsion points of the elliptic curve to this field gives all abelian extensions. If one thinks carefully, then one notes that in the case of $\mathbb{Q}$ as well, there was a geometric object whose torsion points gave rise to abelian extensions. In the case of $\mathbb{Q}$, this was the unit circle, whose torsion points are nothing but roots of unity. Therefore, one hopes for some kind of generalisation to all base fields, which both Kronecker and Hilbert had envisioned.

Shimura was able to extend this to all CM-fields, which are imaginary quadratic extensions of totally real fields. A totally real field is nothing but a number field such that all its embeddings into $\mathbb{C}$ are real. Given any CM-field, one can construct a CM abelian variety associated to it, which acts as a generalisation of a CM elliptic curve, and now adjoining the torsion points of these abelian varieties, one can construct all abelian extensions of the given CM-field. This thesis is aimed towards studying and understanding these constructions.

We will start off with an introduction to the theory of complex multiplication, following Milne and building towards the main theorem of CM. We will see how CM abelian varieties are in correspondence with CM-pairs and study their complex as well as arithmetic theory. We will move on to studying $\mathfrak{a}$-multiplications, which are special kinds of isogenies for these CM abelian varieties, and how the absolute Galois group of the reflex field of a CM field acts on the set of isogeny classes of the associated CM abelian variety via these $\mathfrak{a}$-multiplications, this is the main theorem of CM. Using this main theorem, one can understand the Galois groups of extensions of the reflex field obtained by adjoining values of certain Weber functions at torsion points of the CM abelian variety. Ultimately

these extensions end up being the ray class fields of the reflex field. This is Shimura's work, we merely translate and present his proof in a modern algebro-geometric language.

Chapter 1

Introduction to CM theory

This chapter's aim is to recall and collect definitions and results on CM-algebra and CM-types. We refer to Milne for proofs for the most part, except in some cases where we must supply extra details on our own.

1.1 CM-fields and algebras

CM-fields, as the name suggests, are fields with CM i.e., fields with complex multiplication. For a field to have "complex multiplication" just means that it has a complex conjugation, and we require that this is unique. We formalize this notion now.

Definition 1.1. *A* **totally real field** K *is a finite extension of* $\mathbb{Q}$ *such that all its embeddings* $\rho : K \to \mathbb{C}$ *into* $\mathbb{C}$ *are actually embeddings into* $\mathbb{R}$*, i.e.,* $\mathrm{im}(\rho) \subset \mathbb{R}$ *for all* $\rho : K \hookrightarrow \mathbb{C}$*.*

Definition 1.2. *A* **CM-field** E *is an imaginary quadratic extension of a totally real field* K*.*

One way to see that a field is CM, is to find a non-trivial automorphism on it that acts like the complex conjugation, fixing the largest totally real subfield.

Proposition 1.3 (Milne)**.** *For a number field* E*, the following are equivalent:*

1. E *is a CM-field.*
2. *There exists a unique non trivial automorphism* ι_E *of* E *such that* $\rho \circ \iota_E = \iota \circ \rho$ *for all embeddings* $\rho : E \to \mathbb{C}$*. We call this* ι_E *the complex conjugation on* E*.*
3. $E = K[\alpha]$ *with* K *totally real,* $\alpha^2 \in K$ *and* $\rho(\alpha^2) < 0$ *for all embeddings* $\rho : K \to \mathbb{C}$*.*

Proof. Proposition 1.4 in [3]. □

Corollary 1.3.1. *A finite composition of CM-fields is CM. In particular, the Galois closure of a CM-field in any larger field is CM.*

Proof. Let E_1, E_2 be two CM-fields. Then any embedding of E_1E_2 into $\mathbb{C}$ will induce a complex conjugation on it. This restricts to the unique complex conjugations on E_1 and E_2. Therefore, there is a unique complex conjugation induced on E_1E_2 and we are done by part 2 of 1.3. □

We list some useful facts about CM and totally real fields as just a remark, since they are not so difficult to prove.

Remark 1.4. *1. A composite of totally real fields is totally real.*

2. Any number field contains exactly one largest totally real subfield. Indeed, one can take the composite of all totally real subfields of the number field by item 1.

We can extend the notion of a CM-field to that of a CM-algebra as follows:

Definition 1.5. *A* **CM-algebra** *is a finite product of CM-fields.*

Let E be a CM algebra, then $E = \prod_{j=1}^{n} E_j$. Let ι_j be the complex conjugation on each factor E_j.

Definition 1.6. *Then E has a complex conjugation ι_E defined as*

$$\iota_E(a_1, \ldots, a_n) = (\iota_1(a_1), \ldots, \iota_n(a_n))$$

This ι_E satisfies

$$\iota \circ \rho = \rho \circ \iota_E$$

for all homomorphisms $\rho : E \to \mathbb{C}$ and is the unique such automorphism of E.

Remark 1.7. *1. To highlight the fact that there is a unique complex conjugation, consider the following. We know that E is a product of CM-fields $E = \prod_{i=1}^{n} E_i$. Any embedding of E into $\mathbb{C}$ is a $\mathbb{Q}$-algebra homomorphism $\rho : E \to \mathbb{C}$. On each factor E_i, this embedding is just a field homomorphism $\rho_i : E_i \hookrightarrow E \to \mathbb{C}$. It is easy to see that the complex conjugation on E obtained from pulling back along ρ behaves the same on each E_i as that obtained from pulling back along ρ_i. But the complex conjugation on E_i is the unique non trivial element of $\mathrm{Gal}(E_i/K_i)$ where K_i is the maximal totally real subfield of E_i.*

This means that the complex conjugation on E is unique and does not depend on ρ.

2. What this also means, is that the complex conjugation obtained from an embedding of E into $\bar{\mathbb{Q}}$ is also the same as that obtained from an embedding into $\mathbb{C}$. This is because any complex conjugation on $\bar{\mathbb{Q}}$ is obtained from an embedding of $\bar{\mathbb{Q}}$ into $\mathbb{C}$.

The fixed algebra of ι_E is the product of the largest totally real subfields of the E_j's. Following Milne, we will sometimes write $\bar{a}$ for $\iota_E a$.

1.2 CM-types

Let E be a CM-algebra, and let's fix an algebraic closure of $\mathbb{Q}$ which we call $\bar{\mathbb{Q}}$. The $\mathbb{Q}$-algebra homomorphisms $E \to \bar{\mathbb{Q}}$ occur in complex conjugate pairs. A CM-type on E is nothing but a choice of one embedding out of each conjugate pair.

Definition 1.8. *A* **CM-type** *on a CM-algebra E is a subset $\Phi \subset \mathrm{Hom}(E, \bar{\mathbb{Q}})$ such that*

$$\mathrm{Hom}(E, \bar{\mathbb{Q}}) = \Phi \sqcup \iota\Phi \tag{1.1}$$

for a complex conjugation ι on $\bar{\mathbb{Q}}$.

Note that if (1.1) holds for some complex conjugation, then it holds for all complex conjugations since $\iota\varphi = \varphi\iota_E$ for any embedding $\varphi : E \to \bar{\mathbb{Q}}$, and ι_E is unique due to remark 1.7.

We call the pair (E, Φ) a ***CM-pair***.

Definition 1.9. *Let $F \subset E$ be a CM-subalgebra. We say a CM-type Φ on E is an* **extension** *of a CM-type Φ_F on F iff $\Phi|_F = \Phi_F$, where,*

$$\Phi|_F := \{\varphi|_F \mid \varphi \in \Phi\}$$

Proposition 1.10 (Milne)**.** *Let $F \subset E$ be a CM-subalgebra.*

1. *If Φ_F is a CM-type on F, then there is a unique CM-type Φ on E that is an extension of Φ_F.*
2. *A CM-type Φ on E is an extension of a CM-type on F iff $\Phi|_F$ is a CM-type on F.*

Proof. 1. Define $\Phi := \{\varphi : E \to \bar{\mathbb{Q}} \mid \varphi|_F \in \Phi_F\}$. Note that this set has $[E : \mathbb{Q}]/2$ elements since Φ_F has $[F : \mathbb{Q}]/2$ elements and each $\varphi \in \Phi_F$ has $[E : F]$ extensions to E. Now if $\varphi, \iota\varphi \in \Phi$ for some $\varphi \in \Phi$, then $\varphi|_F, \iota\varphi|_F \in \Phi_F$, which is a contradiction, so Φ is a CM-type on E. Now if Φ' is any extension of Φ_F to E, then each $\varphi \in \Phi_F$ can only have atmost $[E : F]$ extensions and since $|\Phi'| = |\Phi_F| \cdot [E : F]$, it forces each φ to have exactly $[E : F]$ many extensions in Φ', which forces $\Phi' = \Phi$, proving uniqueness.

2. If the restriction of Φ is a CM-type, then automatically its extension is Φ itself by uniqueness guaranteed from part 1.

   The other direction is true because $(\Phi_F)_E|_F = \Phi_F$, where Φ_F is a CM-type on F and $(\Phi_F)_E$ denotes its extension to E.

□

This motivates the following:

Definition 1.11. *A CM pair (E, Φ) is called* **primitive** *if E is a field and there does not exist a subfield $F \subset E$ such that $\Phi|_F$ is a CM-type on F.*

One can also show that any CM-pair arises from a unique primitive pair.

Proposition 1.12 (Milne)**.** *Every CM-pair (E, Φ) with E a field, is the extension of a unique primitive CM-pair (F, Φ_F) with $F \subset E$. Moreover, for any CM-field E' containing E and Galois over $\mathbb{Q}$, F is the fixed field of*

$$H = \{\sigma \in \mathrm{Gal}(E'/\mathbb{Q}) \mid \Phi'\sigma = \Phi'\}$$

Here, Φ' is the extension of Φ_F to E'.

Proof. We claim that it is enough to just treat the case when $E/\mathbb{Q}$ is Galois, in which case, $(F := E^H, \Phi|_F)$ is the unique primitive CM-subpair. To see why this is enough, note that for any CM-subfield $E_0 \subset E$, with CM-type $\Phi_0 = \Phi|_{E_0}$, one has that $F \subset E_0$, which automatically means that Φ_0 is induced by Φ_F. Indeed, let $\sigma \in \mathrm{Gal}(E/E_0)$, we have

$$\Phi_0\sigma = \Phi_0$$

which means that $\Phi\sigma = \Phi$, implying $\sigma \in H$. Therefore, $F \subset E_0$ as desired. Note that this is the only primitive CM-subpair of (E_0, Φ_0) since if there was another, that would contradict the claim that (E, Φ) has a unique primitive CM-subpair.

So assume that E is Galois. Now to show that $(F, \Phi|_F)$ is actually a CM-pair, we show that the complex conjugation ι_E restricts to a non trivial automorphism ι_F of F. This is enough to show that F is a CM-field by (b) of 1.3. Indeed, for all $a \in F$ and $\sigma \in H$, we have

$$\Phi\iota_E\sigma\iota_E = \iota\Phi\sigma\iota_E = \iota\Phi\iota_E = \Phi$$

Therefore, $\iota_E \sigma \iota_E \in H$ giving us,

$$\iota_E \sigma \iota_E(a) = a \implies \sigma(\iota_E a) = \iota_E a$$

hence $\iota_E a \in F$ as desired. This shows that $\iota_E|_F$ is an automorphism of F. It is non trivial because

$$\Phi \iota_E = \iota \Phi \neq \Phi$$

therefore, ι_E is not in H, therefore it acts nontrivially on F.

Now if $\Phi|_F$ is not a CM-type on F, then there are $\sigma, \sigma' \in \Phi$ such that $\sigma \iota_E$ and σ' are equal on F. This implies $\sigma \iota_E \in \sigma' H \subset \Phi$ which is a contradiction to (E, Φ) being a CM-pair. To show that (F, Φ_F) is primitive, note that if there was a primitive subpair $(F', \Phi_{F'})$ of (F, Φ_F), then it would correspond to a subgroup H' of $\mathrm{Gal}(E/\mathbb{Q})$ by Galois theory. This subgroup fixes Φ, hence is a subgroup of H, which means $F' \supset F$, which means $F' = F$.

For uniqueness, note that if there are two primitive CM-subpairs (E_1, Φ_1) and (E_2, Φ_2) of (E, Φ), then they correspond to two subgroups H_1 and H_2 of $\mathrm{Gal}(E/\mathbb{Q})$ via Galois theory. Both these subgroups fix Φ and so the subgroup generated by both of them does as well. Then the subfield corresponding to that forms a smaller primitive CM-subpair of (E_i, Φ_i), contradicting the fact that (E_i, Φ_i) was primitive for each i. □

Corollary 1.12.1 (Milne)**.** *A CM-pair (E, Φ) is primitive iff for some (hence all) CM-fields E' containing E and Galois over $\mathbb{Q}$, the subgroup of $\mathrm{Gal}(E'/\mathbb{Q})$ fixing E is*

$$\{\sigma \in \mathrm{Gal}(E'/\mathbb{Q}) \mid \Phi' \sigma = \Phi'\}$$

where Φ' is the extension of Φ to E'.

1.3 Reflex field of a CM-pair

Our eventual goal is to construct abelian varieties with complex multiplication from a given CM-pair (E, Φ). It turns out that given a CM-pair (E, Φ), one can construct a CM abelian variety with CM by E over a field k, iff k contains the "reflex field" of (E, Φ). This will be explained in more detail in later sections, but for now we give the definition of the reflex field.

Proposition 1.13 (Milne)**.** *Let (E, Φ) be a CM-pair. The following conditions on a subfield E^* of $\bar{\mathbb{Q}}$ are equivalent:*

1. *$\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ fixes E^* iff $\sigma \Phi = \Phi$.*
2. *E^* is the subfield of $\bar{\mathbb{Q}}$ generated by the elements $\sum_{\varphi \in \Phi} \varphi(a),\ a \in E$.*

Proof. Proposition 1.16 of [3]. □

Definition 1.14. *The field satisfying the equivalent conditions in the proposition is called the* **reflex field** *E^* of (E, Φ).*

We have the following important properties of E^*:

Proposition 1.15 (Milne)**.** *Let (E, Φ) be a CM-pair.*

1. *The reflex field E^* is a CM-field.*

2. *If* $(E,\Phi)=\prod_{1\le i\le m}(E_i,\Phi_i)$, *then* $E^*=E_1^*\dots E_m^*$.

3. *The reflex field of any extension of* (E,Φ) *equals that of* (E,Φ).

Proof. Proposition 1.18 of [3]. □

The last part of the proposition says that the reflex field is really an object attached to a primitive CM-pair. In the case when E is a subfield of $\bar{\mathbb{Q}}$, we get another description of E^*.

Proposition 1.16 (Milne)**.** *Let* (E,Φ) *be a CM-pair with* $E\subset\bar{\mathbb{Q}}$. *Let* E' *be the Galois closure of* E *and let* Φ' *be the extension of* Φ *to* E'. *Regard the elements of* Φ' *as automorphisms of* E' *and let* $\Phi'^{-1}:=\{\varphi^{-1}\mid\varphi\in\Phi'\}$. *Then* Φ'^{-1} *is a CM-type on* E' *and the primitive pair* (F,Φ_F) *of* (E',Φ'^{-1}) *has* $F=E^*$. *The pair* (F,Φ_F) *is denoted* (E^*,Φ^*) *and is called the reflex CM-pair of* (E,Φ).

Proof. The proof is simply an application of 1.12. By 1.12, we have $F=E'^H$ where $H:=\{\sigma\in\mathrm{Gal}(E'/\mathbb{Q})\mid\Phi'^{-1}\sigma=\Phi'^{-1}\}$. But this means that

$$H=\{\sigma\in\mathrm{Gal}(E'/\mathbb{Q})\mid\sigma\Phi'=\Phi'\}$$

It is now clear by definition of reflex field, that $F=E'^H$ is the reflex field E^* of E' and hence of E. □

Corollary 1.16.1. *Let* E *be a CM-subfield of* $\mathbb{C}$ *and let* (E,Φ) *be a CM-type on it, then*

$$(E^{**},\Phi^{**})=(F,\Phi_F)$$

where (F,Φ_F) *is the primitive CM-subpair of* (E,Φ).

Proof. The proof is simply an application of 1.16. □

Remark 1.17. *It is worthy to note that for an imaginary quadratic field* K, *the reflex field* K^* *is* K *itself. This is just a quick application of 1.16. This is the reason why reflex fields do not appear in the theory of complex multiplcation for elliptic curves.*

1.4 Reflex norm

Let k be a number field. A finitely generated $E\otimes_{\mathbb{Q}}k$ module is the same as a finite dimensional $\mathbb{Q}$-vector space together with commuting $\mathbb{Q}$-linear actions of E,k.

Proposition 1.18 (Milne)**.** *Let* (E,Φ) *be a CM-pair, and let* k *be a subfield of* $\bar{\mathbb{Q}}$. *There exists a finitely generated* $E\otimes_{\mathbb{Q}}k$*-module* V *such that*

$$\mathrm{Tr}_k(a|_V)=\sum_{\varphi\in\Phi}\varphi(a),\ \forall\ a\in E \tag{1.2}$$

iff $k\supset E^*$, *in which case* V *is unique upto* $E\otimes_{\mathbb{Q}}k$*-isomorphism.*

Note that the trace in (1.2) is being calculated while thinking of V as a k-vector space.

Proof. Proposition 1.21 of [3]. □

Corollary 1.18.1 (Milne)**.** *The reflex field E^* is the smallest subfield of $\bar{\mathbb{Q}}$ such that there exists an $E \otimes_{\mathbb{Q}} E^*$-module V with*

$$V \otimes_{E^*} \bar{\mathbb{Q}} \simeq \bigoplus_{\varphi \in \Phi} \bar{\mathbb{Q}}_\varphi \quad \text{as an } E \otimes_{\mathbb{Q}} \bar{\mathbb{Q}}\text{-module.}$$

where $\bar{\mathbb{Q}}_\varphi$ is a one dimensional $\bar{\mathbb{Q}}$ vector space on which E acts through φ.

Proof. It is clear from the proposition that a finitely generated $E \otimes_{\mathbb{Q}} k$-module V exists such that 1.2 holds. Tensoring with $\bar{\mathbb{Q}}$ over E^* and using uniqueness gives the required isomorphism with $\bigoplus_{\varphi \in \Phi} \bar{\mathbb{Q}}_\varphi$. □

This discussion implies that there exists an almost canonical module associated with a CM-pair E, such that the trace of any element of E that acts on the module is given by 1.2.

Remark 1.19. *It is this canonical module that will turn out to be the tangent space at the origin of the abelian variety with CM by E over k. It is also fruitful to talk about its norm.*

Let V_Φ be an $E \otimes_{\mathbb{Q}} k$ module satisfying 1.2. An element $a \in k$ defines an endomorphism of V_Φ regarded as an E-module, whose determinant is denoted by $\det_E(a|_{V_\Phi})$. If $a \in k^\times$, then $\det(a|_{V_\Phi}) \in E^\times$, so we get a homomorphism

$$N_{k,\Phi} : k^\times \to E^\times$$

More generally, for any $\mathbb{Q}$-algebra R, we have that $V_\Phi \otimes_{\mathbb{Q}} R$ is an $E \otimes_{\mathbb{Q}} R$-module with a $k \otimes_{\mathbb{Q}} R$-action. So we get another norm map,

$$N_{k,\Phi} : (k \otimes_{\mathbb{Q}} R)^\times \to (E \otimes_{\mathbb{Q}} R)^\times$$

This is independent of V_Φ and functorial in R, and is called the ***reflex norm*** from k to E relative to Φ.

Proposition 1.20 (Milne)**.** *For any number field $k \subset \bar{\mathbb{Q}}$ containing E^*,*

$$N_{k,\Phi} = N_\Phi \circ \mathrm{Nm}_{k/E^*} \tag{1.3}$$

where $N_\Phi := N_{E^*,\Phi}$.

Proof. Let V be the $E \otimes_{\mathbb{Q}} E^*$-module satisfying (1.2) and let $V' := V \otimes_{E^*} k$. Let $\{v_j\}_{j=1}^n$ be a basis for V over E, and $\{k_i\}_{i=1}^m$ a basis for k over E^*.

Now, let $a \in k$ act on V' via

$$A := \begin{pmatrix} A_{11} & \dots & A_{1m} \\ A_{21} & \dots & A_{2m} \\ \vdots & \ddots & \vdots \\ A_{m1} & \dots & A_{mm} \end{pmatrix}$$

where each A_{ij} is an $n \times n$ matrix. The matrix A is written with respect to the E-basis $\{k_i \otimes v_j\}_{(i,j)}$ ordered via lexicographic ordering on the pairs (i, j).

Suppose the matrix (a_{ij}) is the matrix representing the action of a on k w.r.t. the E^*-basis $\{k_i\}_{i=1}^m$, we will show that each A_{ij} is the matrix associated to the action of $a_{ij} \in E^*$ on the vector space V, with respect to the E-basis $\{v_j\}_{j=1}^n$. This will be enough to conclude our result, since we can use the following lemma[1]:

[1]Thanks to Christoph for suggesting this lemma.

Lemma 1.21. *Let A be a matrix of the form*

$$\begin{pmatrix} A_{11} & \dots & A_{1m} \\ A_{21} & \dots & A_{2m} \\ \vdots & \ddots & \vdots \\ A_{m1} & \dots & A_{mm} \end{pmatrix}$$

such that each A_{ij} is a square matrix of size n, and such that each of the A_{ij}'s commute with each other. Then we have

$$\det(A) = \det(B)$$

where B is the $n \times n$ matrix obtained by taking the formal determinant of A, thinking of the block matrices A_{ij}'s as entries.

Proof. Refer to [1]. □

Note that when we apply this lemma in our case, B will just be the matrix corresponding to the action of $N_{k/E^*}(a)$ on V and this will finish the proof.

For simplicity, let us look at the action of a on $k_1 \otimes v_j$. On the one hand, we have

$$a(k_1 \otimes v_j) = \sum_{i=1}^{m} \sum_{\ell=1}^{n} A_{i1}^{\ell j}(k_i \otimes v_\ell)$$

where $A_{i1}^{\ell j}$ is the entry in the ℓth row and j^{th} column of A_{i1}.

On the other hand we have

$$a(k_1 \otimes v_j) = (ak_1 \otimes v_j) = \left(\sum_{i=1}^{m} a_{i1} k_i \otimes v_j \right) = \sum_{i=1}^{m} k_i \otimes a_{i1} v_j$$

Therefore we have

$$a_{i1} v_j = \sum_{\ell=1}^{n} A_{i1}^{\ell j} v_\ell$$

Therefore, one sees that A_{i1} acts on V via multiplication by $a_{i1} \in E^*$. With the same logic, one can extend this to all A_{ij}'s. The matrix (a_{ij}) is by definition the one associated to the action of a on k wrt the E^*-basis $\{k_i\}$. And since a_{ij}'s are elements of the commutative algebra E^*, their actions on V via matrices A_{ij} obviously commutes, so we can apply the lemma. □

Lemma 1.22. *From the above definition of $N_{k,\Phi} : (k \otimes_{\mathbb{Q}} R)^\times \to (E \otimes_{\mathbb{Q}} R)^\times$, one can define a continuous homomorphism on the group of idèles:*

$$N_{\mathbb{A}} : \mathbb{A}_k^\times \to \mathbb{A}_E^\times$$

Proof. From $N_{k,\Phi}$, we obtain homomorphisms,

$$N_0 : k^\times \to E^\times$$
$$N_\ell : k_\ell^\times \to E_\ell^\times$$
$$N_\infty : k_\infty^\times \to E_\infty^\times$$

where $A_\ell := A \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ and $A_\infty := A \otimes_{\mathbb{Q}} \mathbb{R}$ for $A = k, E$.

From these maps, we get continuous homomorphisms on the group of idèles $N_{\mathbb{A}} : \mathbb{A}_k^\times \to \mathbb{A}_E^\times$. This is because the matrix through which an element of k_ℓ acts changes continuously with the element, therefore each N_ℓ and N_∞ is continuous.

Note that k_ℓ is a product of local fields, and suppose that π is the prime in one of the factors of k_ℓ and ℓ is the prime in $\mathbb{Q}_\ell$ with $\pi^e = u\ell$ for some $e \in \mathbb{N}$ and some $u \in \mathcal{O}_{k_\ell}^\times$. Then ℓ acts on $V_{\Phi\ell} := V_\Phi \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ by multiplication by ℓ. Therefore, $\det_{\mathbb{Q}_\ell}(\ell) = \ell^n$ for some positive integer n. Thus, $\det_{E_\ell}(\ell) = \ell^m$ for some positive integer m. This gives us $N_\ell(\ell) = \ell^m \implies N_\ell(\pi^e) = \ell^m N_\ell(u)$. If we can show that $N_\ell(u) \in \mathcal{O}_{E_\ell}^\times$, then we will have that $N_\ell(\pi) \in \ell^{m/e}\mathcal{O}_{E_\ell}^\times$, giving us $N_\ell^{-1}(\mathcal{O}_{E_\ell}^\times) = \mathcal{O}_{k_\ell}^\times$, which will mean that the homomorphism induced on the idèles is indeed continuous.

To show that $N_\ell(u) \in \mathcal{O}_{E_\ell}^\times$, note that we only need to show $N_{k,\Phi}(\mathcal{O}_k) \subset \mathcal{O}_E$. Indeed if this is true, then $N_\ell(\mathcal{O}_{k_\ell}) \subset \mathcal{O}_{E_\ell}$ via continuity of N_ℓ, which will imply $N_\ell(\mathcal{O}_{k_\ell}^\times) \subset \mathcal{O}_{E_\ell}^\times$.

So let $a \in \mathcal{O}_k$ act on V_Φ via a matrix $A \in \mathrm{GL}_n(E)$. Since $a \in \mathcal{O}_k$, we have that

$$\sum_{i=0}^{m} \alpha_i a^i = 0$$

with $\alpha_i \in \mathbb{Z}$ and $\alpha_m = 1$. This gives

$$\sum_{i=0}^{m} \alpha_i A^i = 0$$

So A satisfies a polynomial in $\mathbb{Z}[x]$, meaning its minimal polynomial lies in $\mathcal{O}_E[x]$, this follows from Gauss' Lemma about products of primitive polynomials (cf 1.23). Indeed, suppose $m(x)$ is the minimal polynomial of A in $E[x]$. Then $f = g \cdot m$ where $f(x) := \sum_{i=0}^m \alpha_i x^i$ and g is defined as the quotient of f by m in $E[x]$. Now, if $m(x)$ is not in $\mathcal{O}_E[x]$, there are elements $d_1, d_2 \in \mathcal{O}_E[x]$ such that $d_1 m(x)$ and $d_2 g(x)$ are primitive polynomials in $\mathcal{O}_E[x]$. Therefore, we have $d_1 d_2 f(x) = (d_1 m(x))(d_2 g(x))$. But now the content (defined in 1.23) on the left side is $d_1 d_2$, while on the right it is 1 by 1.23. Therefore, $d_1 d_2 = 1$ and d_1, d_2 are units in $\mathcal{O}_E$, hence $m(x), g(x) \in \mathcal{O}_E[x]$. This directly gives us that $\det_E(A) = N_{k,\Phi}(a) \in \mathcal{O}_E$.

Also note that $N_{\mathbb{A}}$ acts via N_0 on the image of $k^\times$ inside $\mathbb{A}_k^\times$. Therefore, this induces a map on the idèle class groups of k and E. □

Lemma 1.23 (Gauss)**.** *Let $f, g \in R[x]$ with R a unital ring. Define by $c(f)$ the "content" of f,*

$$c(f) := \langle a_0, a_1, \ldots, a_n \rangle$$

is the ideal of R generated by the coefficients of f, where $\sum_{i=0}^m a_i x^i = f(x)$. Now if $c(f) = c(g) = R$, then $c(fg) = R$.

Proof. One notes that $c(fg) \subset c(f)c(g)$ always holds.

Now suppose $c(f) = c(g) = R$ and assume $c(fg) \subsetneq R$, which means that there is a maximal ideal $\mathfrak{m}$ of R such that $c(fg) \subset \mathfrak{m} \subsetneq R$. If $f(x) = \sum_{i=0}^m a_i x^i$ and $g(x) = \sum_{j=0}^n b_j x^j$, then

$$c(fg) = \langle \sum_{i+j=k} a_i b_j \rangle_{k=0}^{m+n}$$

Thus, $\sum_{i+j=k} a_i b_j \in \mathfrak{m}$ for all $k = 0, \ldots, m+n$. If $b_j \in \mathfrak{m}$ for all j, then we have a contradiction to the fact that $c(g) = R$. So let ℓ be the smallest integer such that $b_\ell \notin \mathfrak{m}$. Then,

$$\sum_{i+j=\ell} a_i b_j \in \mathfrak{m} \implies a_0 \in \mathfrak{m}$$

In general, by looking at the equations for $i+j=\ell+w$ with $w=0,\dots,n$ inductively, one gets

$$a_i \in \mathfrak{m} \ \forall \ i=0,\dots,n$$

so $c(f)\subset\mathfrak{m}$, which is a contradiction. So $c(fg)=R$, as desired. □

Lemma 1.24 (Milne)**.** *Let (E,Φ) be a CM-pair and let R be a $\mathbb{Q}$-algebra and $N_{k,\Phi}:(k\otimes_{\mathbb{Q}}R)^\times\to(E\otimes_{\mathbb{Q}}R)^\times$ as defined above. Then we have*

$$N_{k,\Phi}(a)\cdot\iota_E(N_{k,\Phi}(a))=\mathrm{Nm}_{k\otimes_{\mathbb{Q}}R/R}(a),\ \forall\ a\in(k\otimes_{\mathbb{Q}}R)^\times$$

Proof. This can be seen through realising that $V:=V_\Phi\oplus\iota V_\Phi$ is a rank 1 free $E\otimes_{\mathbb{Q}}k$-module. Thus, if we let $v_1,\dots,v_n$ be a $\mathbb{Q}$-basis for k, then $1\otimes v_i$ is a E-basis for $E\otimes_{\mathbb{Q}}k\cong V$. Therefore,

$$\det_E(a|_V)=N_{k,\Phi}(a)\cdot\iota_E N_{k,\Phi}(a)=\det_{\mathbb{Q}}(a|_k)=\mathrm{Nm}_{k/\mathbb{Q}}(a)$$

So $1\otimes v_i\otimes 1$ becomes an $E\otimes_{\mathbb{Q}}R$-basis for $V\otimes_{\mathbb{Q}}R$, and $v_i\otimes 1$ an R-basis for $k\otimes_{\mathbb{Q}}R$. This implies that for $a\in k\otimes_{\mathbb{Q}}R$

$$N_{k,\Phi}(a)\cdot\iota_E N_{k,\Phi}(a)=\det_{E\otimes_{\mathbb{Q}}R}(a|_{V\otimes_{\mathbb{Q}}R})=\det_R(a|_{k\otimes_{\mathbb{Q}}R})=\mathrm{Nm}_{k\otimes_{\mathbb{Q}}R/R}(a)$$

□

In certain cases, one can get an explicit description of the norm map as follows:

Proposition 1.25 (Milne)**.** *Let $k\subset\bar{\mathbb{Q}}$ be a finite extension of E^* containing all conjugates of E. For any non zero element a of k,*

$$N_{k,\Phi}(a)=\prod_{\varphi\in\Phi}\varphi^{-1}(\mathrm{Nm}_{k/\varphi E}(a)) \tag{1.4}$$

Proof. For $a\in k^\times$, we have

$$\det_E(a:k_\varphi\to k_\varphi)=\varphi^{-1}(\mathrm{Nm}_{k/\varphi E}a)$$

which implies (1.4) because the module $\oplus_{\varphi\in\Phi}k_\varphi$ is a $E\otimes_{\mathbb{Q}}k$ module that satisfies (1.2), where k_φ is the one dimensional $k-$vector space on which E acts via φ.

□

Corollary 1.25.1 (Milne)**.** *With the notations of 1.25, let $\mathfrak{a}$ be a fractional ideal of k, then*

$$N_{k,\Phi}(\mathfrak{a})=\prod_{\varphi\in\Phi}\varphi^{-1}(\mathrm{Nm}_{k/\varphi E}(\mathfrak{a}))$$

Proof. Note that each side of (1.4) defines a homomorphism on the group of fractional ideals of k to that of E. The quotient q of these two potentially different homomorphisms is again a homomorphism with the same domain and codomain, with trivial image on the principal ideals. This means that q factors through the class group of k, but this is finite, and the codomain is a torsion free group, this means that q has trivial image. Therefore the two sides of (1.4) are equal for all fractional ideals of k. □

Corollary 1.25.2. *Let (E,Φ) be a CM-pair with $E\subset\mathbb{C}$, then using 1.16 we get the reflex pair (E^*,Φ^*). In this case, we get*

$$N_\Phi(a)=\omega(a)\cdot\prod_{\psi\in\Phi^*}\psi(a)\in E$$

for all $a\in E^*$, where $\omega(a)$ is a root of unity depending on the choice of a.

Proof. Let $L \subset \bar{\mathbb{Q}}$ be the Galois closure of E, then we know

$$N_{L,\Phi}(a) = \prod_{\varphi \in \Phi} \varphi^{-1}(\mathrm{Nm}_{L/\varphi E}(a)) = N_\Phi(a^{[L:E^*]}) = N_\Phi(a)^{[L:E^*]}$$

But $\mathrm{Nm}_{L/\varphi E}(a) = \prod_{\psi \in \mathrm{Gal}(L/\varphi E)} \psi(a)$, which gives us

$$N_\Phi(a)^{[L:E^*]} = \prod_{\varphi \in \Phi} \prod_{\psi \in \mathrm{Gal}(L/\varphi E)} \varphi^{-1}\psi(a)$$

Here, we think of elements of Φ as elements of $\mathrm{Gal}(L/E)$ by choosing extensions of each φ to L.

This gives

$$\begin{aligned} N_\Phi(a)^{[L:E^*]} &= \prod_{\varphi \in \Phi} \prod_{\psi \in \mathrm{Gal}(L/\varphi E)} \varphi^{-1}\psi(a) \\ &= \prod_{\varphi \in \Phi_L^{-1}} \varphi(a) \\ &= \prod_{\varphi \in \Phi^*} \varphi(a)^{[L:E^*]} \\ \implies N_\Phi(a) &= \omega(a) \cdot \prod_{\varphi \in \Phi^*} \varphi(a) \end{aligned}$$

for some root of unity $\omega(a)$, as desired. Here we used the fact that any element of Φ_L^{-1} can be written uniquely as $\varphi^{-1}\psi$ with $\varphi \in \Phi$ and $\psi \in \mathrm{Gal}(L/\varphi E)$. To see this, note that any product $\varphi^{-1}\psi$ with $\varphi \in \Phi$ and $\psi \in \mathrm{Gal}(L/\varphi E)$ takes φE to E and as ψ ranges over elements of $\mathrm{Gal}(L/\varphi E)$, $\varphi^{-1}\psi$ ranges over all extensions of φ^{-1} to L.

□

1.5 Positive involutions

Let B be an algebra over a field k. An ***involution*** of B is a k-linear map $b \mapsto b' : B \to B$ such that $(ab)' = b'a'$ and $(b')' = b$ for $a, b \in B$. Note that $1' = 1$ and $c' = c$ for all $c \in k$.

Let $Q \subset \mathbb{R}$ be a subfield. An involution $'$ on a finite dimensional Q-algebra B is said to be ***positive*** if

$$\mathrm{Tr}_{B/Q}(bb') > 0$$

for all $b \in B$. An involution on B is positive iff its linear extension to $B \otimes_Q \mathbb{R}$ is positive.

We will just state the main results about involutions that we will need later, for more details one can refer to [3].

Proposition 1.26 (Milne)**.** *Let B be a finite dimensional Q-algebra. The following conditions on an involution $'$ on B are equivalent:*

1. *B is semisimple and some faithful B-module V admits a positive definite symmetric Q-bilinear form $\langle , \rangle : V \times V \to Q$ such that*

$$\langle bu, v\rangle = \langle u, b'v\rangle \tag{1.5}$$

for all $b \in B, u, v \in V$.

2. every B-module admits a positive definite symmetric Q-bilinear form satisfying (1.5).

3. $'$ is positive.

Proposition 1.27 (Milne)**.** *The CM-algebras are exactly the finite dimensional commutative $\mathbb{Q}$-algebras admitting a (unique) positive involution that acts nontrivially on each factor. This involution is nothing but ι_E as defined in 1.6.*

Proof. These and other related results on involutions can be found in [3], (1.36)-(1.40). □

Chapter 2

Analytic Theory of Abelian Varieties

In this chapter, we will talk about abelian varieties over the complex numbers with the aim to classify CM abelian varieties over the complex numbers upto isomorphism using CM pairs defined in the last chapter. We will assume familiarity with basic facts about complex tori, but we will still state the necessary theorems for completeness.

2.1 Complex Tori

Let Λ be a lattice in a $\mathbb{C}$-vector space V. Then the quotient manifold V/Λ is a complex manifold with a distinguished point that is the coset Λ. Any pointed complex manifold isomorphic to one such is called a ***complex torus***. Equivalently, one can look at the exponential map from the tangent space of a complex manifold $M$with a distinguished point 0:

$$\exp : \mathrm{Tgt}_0(M) \to M$$

If this map realises M as a quotient of $\mathrm{Tgt}_0(M)$ by a lattice, then M is a complex torus.

Proposition 2.1. *Every complex torus is a compact connected complex Lie group. Conversely, if M is a compact connected complex Lie group, then the exponential map realises M as a complex torus.*

Proof. Refer to (2) of §1.1 in [5]. □

The following characterisation of homomorphisms between complex tori is extremely useful. We state them without proof.

Proposition 2.2. *Let $M \simeq V/\Lambda$ and $M' \simeq V'/\Lambda'$ be complex tori.*

1. *A $\mathbb{C}$-linear map $\alpha : V \to V'$ such that $\alpha(\Lambda) \subset \Lambda'$ defines a holomorphic map $M \to M'$ sending 0 to 0, and every holomorphic map $M \to M'$ sending 0 to 0 is of this form for a unique α.*

2. *Every holomorphic map $M \to M'$ sending 0 to 0 is a group homomorphism.*

Proof. Refer to (2.2) and (2.3) of [3]. □

2.2 Riemann Pairs

Definition 2.3. *A* **Riemann pair** *(Λ, J) is a free $\mathbb{Z}$-module of finite rank Λ together with a complex structure J on $\Lambda \otimes_{\mathbb{Z}} \mathbb{R}$. Here, a complex structure J on a real vector space V is just a linear map $J : V \to V$ such that $J^2 = -\mathrm{id}$.*

Note that the complex structure J on a real vector space V of dimension $2n$ makes V into a complex vector space of dimension n, with action of i defined by $i \cdot v := J(v)$ for all $v \in V$.

Definition 2.4. *A homomorphism $\alpha : (\Lambda, J) \to (\Lambda', J')$ of Riemann pairs is a homomorphism $\alpha : \Lambda \to \Lambda'$ of $\mathbb{Z}$-modules such that $\mathrm{id} \otimes \alpha : \mathbb{R} \otimes \Lambda \to \mathbb{R} \otimes \Lambda'$ is a $\mathbb{C}$-linear map.*

Proposition 2.5. *The functor $(\Lambda, J) \mapsto M(\Lambda, J) := (\mathbb{R} \otimes \Lambda, J)/\Lambda$ is an equivalence of categories from the category of Riemann pairs to the category of complex tori.*

Proof. Use 2.2 to see that the functor is fully faithful. It is essentially surjective by definition of complex tori as quotients of $\mathbb{C}^n$ by a lattice. □

2.3 Riemann Forms

Let (Λ, J) be a Riemann pair.

Definition 2.6. *An* **integral Riemann form** *(or just* **Riemann form***) for (Λ, J) is an alternating $\mathbb{Z}$-bilinear form $\psi : \Lambda \times \Lambda \to \mathbb{Z}$ such that*

$$\psi_J : \Lambda_{\mathbb{R}} \times \Lambda_{\mathbb{R}} \to \mathbb{R}$$

$$(x, y) \mapsto \psi_{\mathbb{R}}(x, Jy)$$

is symmetric and positive definite.

The requirement that ψ_J be symmetric and positive definite is the same as saying that it is the imaginary part of a positive definite hermitian form that takes integer values on Λ. A Riemann form ψ is non degenerate (because ψ_J is positive definite), therefore for any $\alpha \in \mathrm{End}^0(\Lambda, J)$, there exists a unique $\alpha' \in \mathrm{End}^0(\Lambda, J)$ such that

$$\psi_{\mathbb{R}}(\alpha x, y) = \psi_{\mathbb{R}}(x, \alpha' y), \text{ for all } x, y \in \Lambda_{\mathbb{R}}$$

The map $\alpha \mapsto \alpha'$ is an involution on $\mathrm{End}^0(\Lambda, J)$, called the ***Rosati involution*** (relative to ψ). For $x, y \in \Lambda_{\mathbb{R}}$ and $\alpha \in \mathrm{End}^0(\Lambda, J)$,

$$\psi_J(\alpha x, y) = \psi_{\mathbb{R}}(\alpha x, Jy) = \psi_{\mathbb{R}}(x, \alpha' Jy) = \psi_{\mathbb{R}}(x, J\alpha' y) = \psi_J(x, \alpha' y)$$

Therefore the Rosati involution is positive by applying 1.26 to the faithful $\mathrm{End}^0(\Lambda, J)$-module $\Lambda_{\mathbb{Q}} := \Lambda \otimes_{\mathbb{Z}} \mathbb{Q}$.

Definition 2.7. *A* **rational Riemann form** *is an alternating $\mathbb{Q}$-bilinear form $\psi : \Lambda_{\mathbb{Q}} \times \Lambda_{\mathbb{Q}} \to \mathbb{Q}$ such that $\psi_J(x, y) := \psi_{\mathbb{R}}(x, Jy)$ is symmetric and positive definite.*

In particular, this means that if ψ is a rational Riemann form, then $\psi(\Lambda, \Lambda) \subset \frac{1}{m}\Lambda$ and that $m\psi$ is an integral riemann form for some integer m.

Remark 2.8 (CM types give Riemann pairs)**.** *Let (E, Φ) be a CM-pair with Λ a lattice inside E. Then Λ embeds into $\mathbb{C}^\Phi$ as a lattice since $E \otimes_{\mathbb{Q}} \mathbb{R} \cong \prod_{\varphi \in \Phi} \mathbb{C} = \mathbb{C}^\Phi$. This means that $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = \mathbb{C}^\Phi$ which has a natural complex structure and so we get a Riemann pair (Λ, J_Φ).*

Here, the isomorphism $E \otimes_{\mathbb{Q}} \mathbb{R} \cong \prod_{\varphi \in \Phi} \mathbb{C}$ is given by

$$e \otimes z \mapsto (\varphi(e) \cdot z)_{\varphi \in \Phi}$$

It is easy to see that this is a $\mathbb{R}$-linear map and that both sides have equal dimension over $\mathbb{R}$.

To see that this is an isomorphism, it is enough to show it is injective. Let $e_1, \ldots, e_{2n}$ be the basis of $E/\mathbb{Q}$. Any element of $E \otimes_{\mathbb{Q}} \mathbb{R}$ is of the form

$$\sum_{i=1}^{2n} e_i \otimes r_i$$

for $r_i \in \mathbb{R}$. Suppose this element maps to 0. Then we define

$$S := \begin{pmatrix} \varphi_1(e_1) & \ldots & \varphi_1(e_{2n}) \\ \vdots & \ddots & \vdots \\ \varphi_n(e_1) & \ldots & \varphi_n(e_{2n}) \end{pmatrix}$$

which gives

$$S \cdot \begin{pmatrix} r_1 \\ \vdots \\ r_{2n} \end{pmatrix} = 0$$

Therefore,

$$\begin{pmatrix} S \\ \bar{S} \end{pmatrix} \cdot \begin{pmatrix} r_1 \\ \vdots \\ r_{2n} \end{pmatrix} = 0$$

and so $\mathrm{disc}_{\mathbb{Q}}(E) = \det\left(\begin{pmatrix} S \\ \bar{S} \end{pmatrix}\right) = 0$ which is a contradiction.

Lemma 2.9 (Milne)**.** *Following the notation as in remark 2.8, we have that $E \subset \mathrm{End}^0(\Lambda, J_\Phi)$ and that the Rosati involution when restricted to E, coincides with the complex conjugation ι_E.*

Proof. Let $R := \{a \in E \mid a\Lambda \subset \Lambda\}$, this is an order in $\mathcal{O}_E$. Indeed, it is a subring of $\mathcal{O}_E$ because any element of R defines a $\mathbb{Z}$-linear map on Λ, therefore has a minimal polynomial that is monic with integer coefficients. If we take any element $a \in E$, and suppose $\Lambda = \mathbb{Z}\langle v_1, \ldots, v_n\rangle$, then there is an integer m such that $mav_i \in \Lambda$ for all i, therefore $ma \in R$, and so $R \otimes_{\mathbb{Z}} \mathbb{Q} = E$, which means R is an order of E.

Since $R \subset \mathrm{End}(\Lambda, J_\Phi)$, therefore $E \subset \mathrm{End}^0(\Lambda, J_\Phi)$. Now note that any Rosati involution on $\mathrm{End}^0(\Lambda, J_\Phi)$ preserves E.

To see this, let $H(x, y) := \psi(x, J_\Phi y) - i\psi(x, y)$ be a Hermitian form on $\mathbb{C}^\Phi$, and note that any Rosati involution for ψ is also one for H. Let $x, y \in \Lambda$ and $a \in E$. Then under the embedding Φ, they are represented by vectors $\vec{x}, \vec{y}, \vec{a} \in \mathbb{C}^\Phi$. The action of $\vec{a}$ on $\Phi(\Lambda)$ is given by componentwise multiplication of the vector entries. Therefore,

$$H(\vec{a}\vec{x}, \vec{y}) = (\vec{a}\vec{x})^t A\overline{(\vec{y})} = (\vec{x})^t A\overline{(\bar{\vec{a}}\vec{y})} = H(\vec{x}, \bar{\vec{a}}\vec{y})$$

Therefore, any Rosati involution acts via the unique complex conjugation ι_E on E. □

Lemma 2.10 (Milne)**.** *Following the notation of remark 2.8, let $\psi : E \times E \to \mathbb{Q}$ be a rational Riemann form, then there exists an element $\alpha \in E^\times$, such that*

$$\psi(x, y) = \mathrm{Tr}_{E/\mathbb{Q}}(\alpha x \bar{y})$$

for all $x, y \in E$. This α satisfies $\bar{\alpha} = -\alpha$ and $\mathrm{Im}(\varphi(\alpha)) > 0$ for all $\varphi \in \Phi$.

Proof. We want to look for a rational Riemann form on (Λ, J_Φ), which is the same as finding a rational bilinear form on E: $\psi : E \times E \to \mathbb{Q}$ such that

1. $\psi(ax, y) = \psi(x, \bar{a}y)\ \forall\ x, y, a \in E$. This is due to lemma 2.9.
2. $\psi(x, y) = -\psi(y, x)\ \forall\ x, y \in E$.
3. $\psi(J_\Phi x, J_\Phi y) = \psi(x, y)$, for all $x, y \in E \otimes_{\mathbb{Q}} \mathbb{R}$. This is equivalent to ψ_{J_Φ} being symmetric.
4. $\psi(x, J_\Phi x) > 0$ for all nonzero $x \in E \otimes_{\mathbb{Q}} \mathbb{R}$. This is equivalent to ψ_{J_Φ} being positive definite.

Define $E^\vee := \{f : E \to \mathbb{Q} \mid f \text{ is } \mathbb{Q}\text{-linear.}\}$ We first prove that the map $E \to E^\vee$ given by:

$$e \mapsto (x \mapsto \mathrm{Tr}_{E/\mathbb{Q}}(ex))$$

is an isomorphism of $\mathbb{Q}$-vector spaces. We know that E and $E^\vee$ have the same finite dimension over $\mathbb{Q}$, so it's enough to show that E injects into $E^\vee$ with this map. But we know that $x \mapsto \mathrm{Tr}_{E/\mathbb{Q}}(ex)$ is the zero map iff $e = 0$ (since $E/\mathbb{Q}$ is separable). Therefore the linear form $\psi(x, \cdot) : E \to \mathbb{Q}$ is of the form

$$\psi(x, y) = \mathrm{Tr}_{E/\mathbb{Q}}(\alpha(x) \cdot y)$$

where $\alpha(x) \in E$ depends on x. Now we know that $\psi(ax, y) = \psi(x, \bar{a}y)$ which implies that

$$\mathrm{Tr}_{E/\mathbb{Q}}(\alpha(ax) \cdot y) = \mathrm{Tr}_{E/\mathbb{Q}}(\alpha(x) \cdot \bar{a}y)$$

this gives

$$\alpha(ax) = \bar{a}\alpha(x)$$

for all $a, x \in E$. Therefore we see that $\alpha(x) = \bar{x}\alpha(1)$. Denote $\alpha(1)$ by simply α and we get

$$\psi(x, y) = \mathrm{Tr}_{E/\mathbb{Q}}(\alpha\bar{x}y)$$

But since $\mathrm{Tr}_{E/\mathbb{Q}}$ is invariant under complex conjugation, we get

$$\psi(x, y) = \mathrm{Tr}_{E/\mathbb{Q}}(\alpha x\bar{y})$$

where we replaced $\bar{\alpha}$ by α.

So we know that any $\mathbb{Q}$-bilinear form on E satisfying item 1 above, is of the form $(x, y) \mapsto \mathrm{Tr}_{E/\mathbb{Q}}(\alpha x\bar{y})$. One notes that item 2 is equivalent to $\bar{\alpha} = -\alpha$, item 3 holds automatically for any α, and that item 4 is equivalent to $\mathrm{Im}(\varphi(\alpha)) > 0\ \forall\ \varphi \in \Phi$. Indeed, for ψ_{J_Φ} to be positive definite, it is enough that $\psi(x, J_\Phi x) > 0$ for all $x \in E \subset \mathbb{C}^\Phi$. One notes that

$$\begin{aligned}\psi(x, y) &= \mathrm{Tr}_{E/\mathbb{Q}}(\alpha x\bar{y}) \\ &= \sum_{\varphi \in \Phi} \mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(\varphi(\alpha)\varphi(x)\overline{\varphi(y)})\end{aligned}$$

Which implies,

$$\begin{aligned}\psi(x, J_\Phi x) &= \sum_{\varphi \in \Phi} \mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(\varphi(\alpha)\varphi(x)\overline{\varphi(ix)}) \\ &= \sum_{\varphi \in \Phi} \mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(-i\varphi(\alpha)\varphi(x)\overline{\varphi(x)}) \\ &= \sum_{\varphi \in \Phi} \mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(-i\varphi(\alpha))\varphi(x)\overline{\varphi(x)}\end{aligned}$$

Since $\varphi(x)\overline{\varphi(x)} > 0$ always, and one can choose $x \in E$ such that $\varphi'(x)$ is as close to zero as we want for all $\varphi' \neq \varphi \in \Phi$, where φ is a chosen fixed element in Φ, while keeping $\varphi(x)$ as close to 1 as we want, using the approximation theorem. We see that we must have

$$\mathrm{Tr}_{\mathbb{C}/\mathbb{R}}(-i\varphi(\alpha)) > 0 \ \forall \ \varphi \in \Phi$$

This is the same as $\mathrm{Im}(\varphi(\alpha)) > 0$ for all $\varphi \in \Phi$.

□

2.4 Abelian Varieties with CM

Definition 2.11. *We define an abelian variety to be a complex torus that admits a Riemann form.*

A Riemann form in this context is sometimes called a ***polarization*** because it gives rise to an isogeny from A to its dual $A^\vee$ as described below. Let $A \simeq V/\Lambda$ be an abelian variety, then so is the dual abelian variety $A^\vee := V'/\Lambda'$ where V' is the space of $\mathbb{R}$-linear maps $V \to \mathbb{R}$ and $\Lambda' = \{f \in V^\vee \mid f(\Lambda) \subset \mathbb{Z}\}$. Now, an integral Riemann form defines an isogeny from $A \to A^\vee$ given by:

$$[a] \mapsto ([x] \mapsto [\psi(a, x)])$$

where $[a] := a + \Lambda \in A$.

Definition 2.12. *Let B be a semisimple k-algebra for a field k, then we know that*

$$B = \prod_{i=1}^{m} M_{n_i}(D_i)$$

with D_i division algebras over k. Then the reduced degree of B/k is defined as:

$$[B : k]_{\mathrm{red}} := \sum_{i=1}^{m} n_i \cdot [D_i : k_i]^{1/2} \cdot [k_i : k]$$

where for all i, k_i is the centre of D_i.

Proposition 2.13 (Milne)**.** *For any abelian variety A,*

$$2 \dim A \geq [\mathrm{End}^0(A) : \mathbb{Q}]_{\mathrm{red}}$$

When equality holds, $\mathrm{End}^0(\mathrm{A})$ *is a product of matrix algebras over fields.*

Proof. Note that $\mathrm{End}^0(A)$ is a semisimple $\mathbb{Q}$-algebra acting faithfully on the $2 \dim A$-dimensional $\mathbb{Q}$-vector space $H_1(A, \mathbb{Q})$. Indeed, $H_1(A, \mathbb{Q}) = \Lambda \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\mathrm{End}(A)$ acts faithfully on Λ. Now the result follows from the following general lemma. □

Lemma 2.14 (Milne)**.** *Let B be a semisimple k-algebra for a field k. For any faithful B-module M,*

$$\dim_k M \geq [B : k]_{\mathrm{red}}$$

and there exists a faithful module for which equality holds iff the simple factors of B are matrix algebras over their centres.

Proof. Refer to [3] proposition (1.2). Note that in the case when B is a commutative algebra, the faithful module is unique and is B itself. □

Definition 2.15. *A complex abelian variety A is said to have* **complex multiplication** *or* **CM** *if*

$$2\dim A = [\mathrm{End}^0(A) : \mathbb{Q}]_{\mathrm{red}}$$

Proposition 2.16 (Milne)**.** *The following conditions on an abelian variety A are equivalent:*

1. *A has complex multiplication.*
2. *$\mathrm{End}^0(A)$ contains a maximal étale algebra of degree $2\dim A$ over $\mathbb{Q}$.*

Proof. This follows from the lemma 2.17 below. □

Lemma 2.17 (Milne)**.** *Let B be a semisimple k-algebra. Every maximal étale k-subalgebra of B has degree $[B : k]_{\mathrm{red}}$ over k.*

Proof. Propositon 1.3 of [3]. □

Definition 2.18. *Let A be an abelian variety with CM and let E be a maximal étale subalgebra of $\mathrm{End}^0(A)$ of degree $2\dim A$ over $\mathbb{Q}$. In this case we say that A* **has CM by** *E.*

The following proposition shows that one can construct an abelian variety with CM using a CM-pair (E, Φ).

Proposition 2.19. *In 2.8, we saw that given any CM-pair (E, Φ), we can get a Riemann pair (Λ, J_Φ). This leads to a complex torus given by $A_\Phi := \mathbb{C}^\Phi/\Lambda$. We claim that this is actually an abelian variety.*

Proof. To show that A_Φ is an abelian variety, we must show that there exists a Riemann form on $\mathbb{C}^\Phi$, that is a Riemann form on (Λ, J_Φ). This is equivalent to having a rational Riemann form on E. By 2.10, we just need to find an element $\alpha \in E^\times$ such that

$$-\alpha = \bar{\alpha}, \text{ and } \mathrm{Im}(\varphi(\alpha)) > 0 \ \forall\ \varphi \in \Phi$$

Let F be the product of the largest totally real subfields of all factors of E. Then $E = F[\alpha]$ with $\alpha^2 \in F$ which implies $\bar{\alpha} = -\alpha$. The weak approximation theorem implies that α can be chosen so that $\mathrm{Im}(\varphi(\alpha)) > 0$ for all $\varphi \in \Phi$. Therefore, α with the required properties exists, and so (Λ, J_Φ) is polarizable. □

Lemma 2.20. *Let $\psi_1, \psi_2 : E \times E \to \mathbb{Q}$ be two rational Riemann forms defined by*

$$\psi_i(x, y) := \mathrm{Tr}_{E/\mathbb{Q}}(t_i x \bar{y})$$

where $t_i \in E^\times$ such that $\bar{t}_i = -t_i$ and $\mathrm{Im}(\varphi(t_i)) > 0$ for all $\varphi \in \Phi$ (this is due to 2.19)

Then $t_1/t_2 \in F_{\gg 0}$, where $F_{\gg 0}$ denotes the totally positive elements of F.

Proof. We know that $\bar{t}_1/\bar{t}_2 = -t_1/-t_2 = t_1/t_2$, so $t_1/t_2 \in F$. Now since $\mathrm{Im}(\varphi(t_i)) > 0$ for all $\varphi \in \Phi$, so for $\varphi \in \Phi$, $\varphi(t_1/t_2) = \varphi(t_1)/\varphi(t_2) = \mathrm{Im}(\varphi(t_1))/\mathrm{Im}(\varphi(t_2)) > 0$ where the last equality is true because $\varphi(t_i)$ is purely imaginary because $\bar{t}_i = -t_i$. But $\Phi|_F$ are exactly all the embeddings of F, so we are done. □

Remark 2.21. *Let A be an abelian variety isogenous to $\prod_i A_i^{n_i}$ of* ***isotypic abelian varieties*** *(an isotypic abelian variety is by definition a power of a simple abelian variety, in particular this means that A_i is simple for all i). Then $D_i := \mathrm{End}^0(A_i)$ is a division algebra, and $\mathrm{End}^0(A) \simeq \prod_i M_{n_i}(D_i)$. From 2.14, and the fact that $H_1(A, \mathbb{Q})$ is the required faithful module of dimension $2\dim A$, we get that A has CM iff D_i is a field of degree $2\dim A_i$ for all i.*

In particular, a simple abelian variety A has CM iff $\mathrm{End}^0(A)$ is a CM-field of degree $2\dim A$ over $\mathbb{Q}$, and an arbitrary abelian variety has CM iff each simple isogeny factor does.

Remark 2.22. *An abelian variety A with CM by E is isogenous to a principal abelian variety, i.e., an abelian variety on which the whole ring of integers of E acts. One can see this by noting that for any lattices $\mathfrak{a} \subset \mathfrak{b}$ inside E, we get an isogeny $\mathbb{C}^\Phi/\Phi(\mathfrak{b}) \to \mathbb{C}^\Phi/\Phi(\mathfrak{a})$. Now for any lattice Λ inside E, there is a $\mathbb{Z}$-linear map $\alpha : \Lambda \to \Lambda$ with image inside $\mathcal{O}_E$, therefore we have the $\mathbb{C}^\Phi/\Phi(\Lambda) \sim \mathbb{C}^\Phi/\Phi(\alpha(\Lambda)) \sim \mathbb{C}^\Phi/\Phi(\mathcal{O}_E)$. Obviously $\mathbb{C}^\Phi/\Phi(\mathcal{O}_E)$ is principal.*

2.5 Classification of CM Abelian Varieties upto isogeny

Let A be an abelian variety with CM by $i : E \hookrightarrow \mathrm{End}^0(A)$ (refer to 2.18). Note that since $\mathrm{End}^0(A)$ acts faithfully on $H_1(A, \mathbb{Q})$, then E does as well, and applying 2.14 we see that $H_1(A, \mathbb{Q})$ is a rank 1 free E-module. This implies that $H_1(A, \mathbb{C}) = H_1(A, \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C} = \prod_{\varphi : E \to \mathbb{C}} \mathbb{C}$ where $\varphi : E \to \mathbb{C}$ runs over all embeddings of E into $\mathbb{C}$. But we know that

$$H^1(A, \mathbb{C}) = H^0(A, \Omega^1) \oplus \overline{H^0(A, \Omega^1)}$$

here Ω^1 is the sheaf of complex holomorphic 1-forms on A. Note that the global forms on A are exactly the ones that are invariant under the group action of A (refer to §2.7, Proposition 4 of [6]), which means that $H^0(A, \Omega^1) \cong \Omega^1_0 \cong \mathrm{Hom}(\mathrm{T}_0(A), \mathbb{C})$. So by taking duals,

$$H_1(A, \mathbb{C}) = T_0(A) \oplus \overline{T_0(A)}$$

where, $T_0(A)$ is the tangent space of A at 0, and Ω^1_0 is the stalk of Ω^1 at 0. This means that there is a CM-type Φ on E, such that $T_0(A) = \prod_{\varphi \in \Phi} \mathbb{C}_\varphi$ where $\mathbb{C}_\varphi$ is the one dimensional $\mathbb{C}$-vector space on which E acts via $\varphi : E \to \mathbb{C}$. Then we say that A, together with the injective homomorphism $i : E \hookrightarrow \mathrm{End}^0(A)$, is of CM-type (E, Φ).

Let e be a basis vector for $H_1(A, \mathbb{Q})$ as an E-module, and let $\mathfrak{a}$ be a lattice in E such that $\mathfrak{a}e = H_1(A, \mathbb{Z})$. So we get the commutative diagram:

$$\begin{array}{ccc} T_0(A)/\Lambda & \xrightarrow{\cong} & A \\ \downarrow \cong & & \downarrow \\ \mathbb{C}^\Phi/\Phi(\mathfrak{a}) & = & A_\Phi \end{array} \tag{2.1}$$

where Λ is the lattice $H_1(A, \mathbb{Z})$ and the top map is the canonical parametrization of the abelian variety A.

Definition 2.23. *Let (A_1, i_1) and (A_2, i_2) be two abelian varieties with CM by E, given by $i_1 : E \hookrightarrow \mathrm{End}^0(A_1)$ and $i_2 : E \hookrightarrow \mathrm{End}^0(A_2)$. Then we say that they are E-isogenous if there exists an isogeny $\psi : A_1 \to A_2$ such that*

$$\psi(i_1(a) \cdot x) = i_2(a) \cdot \psi(x)$$

for all $a \in E$ and $x \in A$.

Proposition 2.24. *The map $(A, i) \mapsto (E, \Phi)$ constructed above gives a bijection from the set of E-isogeny classes of pairs (A, i) to the set of isomorphism classes of CM-pairs, with inverse $(E, \Phi) \mapsto (A_\Phi, i_\Phi)$ as constructed in 2.19.*

Proof. Given any CM-pair (E, Φ), we have defined an abelian variety (A_Φ, i_Φ), which required a choice of a lattice $\Lambda \subset E$. But we showed in 2.22 that any such variety is isogenous to the principal abelian variety $\mathbb{C}^\Phi/\Phi(\mathcal{O}_E)$, therefore (A_Φ, i_Φ) is well defined upto isogeny. Given an isomorphic CM-pair (E', Φ'), with isomorphism $\alpha : (E, \Phi) \to (E', \Phi')$, and a lattice $\Lambda \subset E$, we get the obvious isomorphism $\alpha : \mathbb{C}^\Phi \to \mathbb{C}^{\Phi'}$ which in fact induces an isomorphism $\alpha : \mathbb{C}^\Phi/\Phi(\Lambda) \to \mathbb{C}^{\Phi'}/\Phi'(\alpha(\Lambda))$.

We also have the following commutative diagram which shows that the isomorphism is actually of pairs (A_Φ, i_Φ) and $(A_{\Phi'}, i_{\Phi'})$:

$$\begin{array}{ccc} \mathbb{C}^\Phi/\Phi(\Lambda) & \xrightarrow{\alpha} & \mathbb{C}^{\Phi'}/\Phi'(\alpha(\Lambda)) \\ {\scriptstyle \Phi(e)}\downarrow & & \downarrow{\scriptstyle \Phi'(e)} \\ \mathbb{C}^\Phi/\Phi(\Lambda) & \xrightarrow[\alpha]{} & \mathbb{C}^{\Phi'}/\Phi'(\alpha(\Lambda)) \end{array}$$

So the map is well defined in at least one direction.

For the other direction, we have the construction as described before the proposition. We need to show that an isogeny $\alpha : (A, i) \to (A', i')$ induces an isomorphism between $(E, \Phi) \to (E', \Phi')$. It is easy to see that α induces an isomorphism between the endomorphism algebras, and since it is must respect the inclusion maps i, i', it induces an isomorphism between E, E'.

The isogeny α also induces a $\mathbb{C}$-linear map $\alpha : \mathbb{C}^\Phi \to \mathbb{C}^{\Phi'}$ such that $\alpha(\Phi(\mathfrak{a})) \subset \Phi'(\mathfrak{a}')$ where the notation is the same as in the paragraph preceding the proposition. The fact that α induces an isomorphism between endomorphism algebras and sends E to E' means that the following diagram commutes:

$$\begin{array}{ccc} \mathbb{C}^\Phi & \xrightarrow{\alpha} & \mathbb{C}^{\Phi'} \\ \uparrow & & \uparrow \\ E & \xrightarrow[\alpha]{} & E' \end{array}$$

where the vertical maps are simply the inclusions $E \hookrightarrow \mathbb{C}^\Phi$ given by $e \mapsto (\varphi(e))_{\varphi \in \Phi}$. This is enough to conclude that $(E, \Phi) \cong (E', \Phi')$ via α.

Now we need to show that these two maps are inverses of each other. Let (E, Φ) be the CM-type of (A, i), then (2.1) shows that $(A, i) \sim (A_\Phi, i_\Phi)$. In the other direction, it is obvious that (A_Φ, i_Φ) is of CM-type (E, Φ) by construction. □

2.6 Classification of CM Abelian Varieties upto isomorphism

Let A be a complex abelian variety of type (E, Φ) with $i : E \hookrightarrow \mathrm{End}^0(A)$, giving the action of E on A. Pick $0 \neq e \in H_1(A, \mathbb{Q})$, this is an E-basis for $H_1(A, \mathbb{Q})$. Then there is a lattice $\mathfrak{a} \subset E$ such that $\mathfrak{a}e = H_1(A, \mathbb{Z})$. We know that this choice of e gives us an isomorphism

$$E/\mathfrak{a} \xrightarrow{\cong} H_1(A, \mathbb{Q})/H_1(A, \mathbb{Z})$$

which in turn gives us an E-isomorphism

$$\theta : \mathbb{C}^\Phi/\Phi(\mathfrak{a}) \xrightarrow{\cong} T_0(A)/\Lambda \cong A$$

Conversely, any E-isomorphism $\mathbb{C}^\Phi/\Phi(\mathfrak{a}) \to A$ arises in this way from an E-basis element $e \in H_1(A, \mathbb{Q})$. If e is replaced by $ae \in E$ with $a \in E^\times$, then θ gets replaced by $\theta \circ a$ and $\mathfrak{a}$ gets replaced by $a^{-1}\mathfrak{a}$.

Now we can classify triples (A, i, ψ) of CM-abelian varieties with polarisation ψ.

Proposition 2.25. *Let (A, i, ψ) be a CM-abelian variety of type (E, Φ) with polarisation ψ. Then any choice of $e \in H_1(A, \mathbb{Q})$ gives us a lattice $\mathfrak{a} \subset E$ such that $a\mathfrak{a} = H_1(A, \mathbb{Z})$, an E-isomorphism $\theta : \mathbb{C}^\Phi/\Phi(\mathfrak{a}) \xrightarrow{\cong} A$ and an element $t \in E$ such that*

$$\psi(ex, ey) = \mathrm{Tr}_{E/\mathbb{Q}}(tx\bar{y}) \ \forall \ x, y \in E$$

*We say that (A, i, ψ) is of **type** $(E, \Phi, \mathfrak{a}, t)$ relative to θ.*

If e is replaced by $a^{-1}e$ with $a \in E^\times$, then θ is replaced by $\theta \circ a^{-1}$ and t is replaced by $t/a\bar{a}$, while $\mathfrak{a}$ is replaced by $a\mathfrak{a}$. So the type of (A, i, ψ) w.r.t. $\theta \circ a^{-1}$ becomes $(E, \Phi, a\mathfrak{a}, t/a\bar{a})$.

The type $(E, \Phi, \mathfrak{a}, t)$ determines (A, i, ψ) upto isomorphism, and (A, i, ψ) determines the type upto a base change of the above form.

Proof. The fact that (A, i, ψ) determines the type upto the above change of base, is obvious from the preceding discussion. For the other direction, one notes that given a CM abelian variety (A, i, ψ) of type $(E, \Phi, \mathfrak{a}, t)$, we have an E-isomorphism

$$\mathbb{C}^\Phi/\Phi(\mathfrak{a}) \xrightarrow{\cong} T_0(A)/\Lambda$$

such that $\psi(ex, ey) = \mathrm{Tr}_{E/\mathbb{Q}}(tx\bar{y}) \ \forall \ x, y \in E$. So we get an E-isomorphism

$$(\mathbb{C}^\Phi/\Phi(\mathfrak{a}), i_\Phi, (x, y) \mapsto \mathrm{Tr}_{E/\mathbb{Q}}(tx\bar{y})) \to (A, i, \psi)$$

□

Chapter 3

Arithmetic Theory of Abelian Varieties

In this chapter, we treat the theory of CM abelian varities over general fields, but our main focus will be on the characteristic 0 case, and in particular, on subfields of $\mathbb{C}$.

3.1 CM abelian varieties over general fields

We recall the definition of an abelian variety over a general field k.

Definition 3.1. *An abelian variety A over a field k, is a complete and connected algebraic group over k.*

Now let A be an abelian variety over a field k. If k embeds into $\mathbb{C}$, then $\mathrm{End}^0(A)$ acts faithfully on $H_1(A(\mathbb{C}), \mathbb{Q})$ which has dimension $2 \dim A$. This implies

$$[\mathrm{End}^0(A) : \mathbb{Q}]_{\mathrm{red}} \leq 2 \dim A$$

due to 2.14. In general, for $\ell \neq \mathrm{char}(k)$, $\mathrm{End}^0(A) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ acts faithfully on $V_\ell A := T_\ell A \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ where $T_\ell A$ is the ℓ-adic Tate module of A. We again have $\dim_{\mathbb{Q}_\ell}(V_\ell A) = 2 \dim A$, therefore by 2.14 we get

$$[\mathrm{End}^0(A) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell : \mathbb{Q}_\ell]_{\mathrm{red}} \leq 2 \dim A$$

but the left hand side is the same as $[\mathrm{End}^0(A) : \mathbb{Q}]_{\mathrm{red}}$. So we get

$$[\mathrm{End}^0(A) : \mathbb{Q}]_{\mathrm{red}} \leq 2 \dim A$$

So we can define, as we did in 2.15:

Definition 3.2. *An abelian variety A over a field k is said to have complex multiplication iff*

$$[\mathrm{End}^0(A) : \mathbb{Q}]_{red} = 2 \dim A$$

Now let E be an étale $\mathbb{Q}$-subalgebra of $\mathrm{End}^0(A)$, then

$$[E : \mathbb{Q}] \leq [\mathrm{End}^0(A) : \mathbb{Q}]_{\mathrm{red}} \leq 2 \dim A$$

where the first inequality holds because of 2.17.

Definition 3.3. *We say that an abelian variety A over a field k has CM by an étale $\mathbb{Q}$-algebra E iff there is an injective map of algebras $i : E \to \mathrm{End}^0(A)$ such that*

$$[E : \mathbb{Q}] = [\mathrm{End}^0(A) : \mathbb{Q}]_{red} = 2 \dim A$$

and in this case, E is a maximal étale $\mathbb{Q}$-subalgebra of $\mathrm{End}^0(A)$.

Proposition 3.4 (Milne)**.** *Let $R := E \cap \mathrm{End}(A)$, then the commutants of R in $\mathrm{End}_{\mathbb{Q}_\ell}(V_\ell)$, $\mathrm{End}_{\mathbb{Z}_\ell}(T_\ell A)$, $\mathrm{End}^0(A)$, $\mathrm{End}(A)$ are E_ℓ, R_ℓ, E, R respectively.*

Proof. Refer to [3], Corollary 7.4. □

3.2 Tate Modules of CM abelian varieties

Let us collect some results on the structure of Tate modules of CM abelian varieties to use later.

Let A be an abelian variety with CM by E over a characteristic 0 field k. Let $\Phi \subset \mathrm{Hom}(E, \bar{\mathbb{Q}})$ be the CM-type of A. We define the **ℓ-adic Tate module** of A as:

$$T_\ell(A) := \varprojlim A_\ell(\bar{k})$$

Let $\dim A = n$, then $T_\ell A$ is a free $\mathbb{Z}_\ell$-module of rank $2n$. We define the **complete Tate module**

$$T_f A := \prod_\ell T_\ell A$$

We define $V_f A := T_f A \otimes_{\mathbb{Z}} \mathbb{Q}$ and $T_f A := \prod_\ell T_\ell A$, where the product ranges over all primes $\ell \in \mathbb{Z}$, and $T_\ell A$ is the ℓ-adic Tate module. Note that $V_\ell A$ is a faithful $E_\ell := E \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$-module of dimension $2 \dim A$ over $\mathbb{Q}_\ell$, so by 2.14 we get that $V_\ell A$ is a free E_ℓ-module of rank 1.

Lemma 3.5. *$V_f A$ is a rank 1 module over $\mathbb{A}_{f,E}$.*

Proof. Consider the exact sequence

$$0 \to R \to \mathcal{O}_E \to \mathcal{O}_E/R \to 0$$

where $R := E \cap \mathrm{End} A$. For all primes $\ell \nmid [\mathcal{O}_E : R]$, we have

$$0 \to R_\ell \to \mathcal{O}_{E,\ell} \to 0$$

where $R_\ell := R \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ and $\mathcal{O}_{E,\ell} := \mathcal{O}_E \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. The sequence remains exact because $\mathbb{Z}_\ell$ is a flat $\mathbb{Z}$-module. $(\mathcal{O}_E/R) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell = 0$ because multiplication by ℓ is an isomorphism on $\mathcal{O}_E/R$, so any element $a \otimes b$ is equal to $a\ell^{-1} \otimes b\ell$. ℓ has a finite order as an automorphism of the finite group $\mathcal{O}_E/R$, so

$$a\ell^{-n} \otimes b\ell^n = a \otimes b\ell^n \implies a(1 - \ell^{-n}) \otimes b\ell^n = 0 \implies a \otimes b(\ell^n - 1) = 0$$

But $\ell^n - 1$ is a unit in $\mathbb{Z}_\ell$, so one sees by replacing b with $b(\ell^n - 1)^{-1}$, that $a \otimes b = 0$. Therefore, $R_\ell \cong \mathcal{O}_{E,\ell}$ which is a product of discrete valuation rings. Therefore, any R_ℓ-module is a product $\prod_i M_i^{m_i}$ of simple modules M_i over R_ℓ. Therefore,

$$T_\ell A = \prod_i M_i^{m_i}$$

for some $m_i \in \mathbb{Z}_{\geq 0}$. $T_\ell A$ is a faithful R_ℓ-module, therefore $m_i \geq 1$ for all i. In addition,

$$T_\ell A \otimes_{\mathbb{Z}} \mathbb{Q} = V_\ell A \cong E_\ell$$

which means that $\mathrm{rank}_{\mathbb{Z}_\ell} T_\ell A = 2 \dim A = \mathrm{rank}_{\mathbb{Z}_\ell} R_\ell$, implying

$$T_\ell A = \prod_i M_i \cong R_\ell$$

for $\ell \nmid [\mathcal{O}_E : R]$. Here we used the Chinese Remainder Theorem to say that $\prod_i M_i \cong R_\ell$ since each $M_i := R_\ell/\mathfrak{m}_i$ for some maximal ideal $\mathfrak{m}_i$ of R_ℓ. Therefore,

$$\begin{aligned} V_f A := T_f A \otimes_{\mathbb{Z}} \mathbb{Q} &\cong \prod_{\ell} T_\ell A \otimes_{\mathbb{Z}} \mathbb{Q} \\ &\cong \left(\prod_{\ell \nmid [\mathcal{O}_E:R]} R_\ell \prod_{\ell \mid [\mathcal{O}_E:R]} T_\ell A \right) \otimes_{\mathbb{Z}} \mathbb{Q} \\ &\cong \left(\prod R_\ell \right) \otimes_{\mathbb{Z}} \mathbb{Q} \\ &\cong \mathbb{A}_{f,E} \end{aligned}$$

where the second last isomorphism holds because $R_\ell \otimes_{\mathbb{Z}} \mathbb{Q} \cong T_\ell A \otimes_{\mathbb{Z}} \mathbb{Q} \cong E_\ell$.

This shows that $V_f A$ is a rank 1 $\mathbb{A}_{f,E}$-module. □

3.3 $\mathfrak{a}$-multiplications

Suppose we have two CM abelian varieties A_1, A_2 of type (E, Φ) such that they have multiplication by $\mathcal{O}_E$ (in this case they are called principal CM) and such that they are isomorphic to $\mathbb{C}^\Phi/\Phi(\mathfrak{a}_1)$ and $\mathbb{C}^\Phi/\Phi(\mathfrak{a}_2)$ respectively, where $\mathfrak{a}_1, \mathfrak{a}_2$ are lattices in E. Then there is an element $a \in \mathbb{Z} \setminus \{0\}$ such that $a\mathfrak{a}_1 \subset \mathfrak{a}_2$. Indeed, if $\{v_i\}_{i=1}^n$ is a $\mathbb{Z}$-basis for $\mathfrak{a}_2$, then any element of a $\mathbb{Z}$-basis for $\mathfrak{a}_1$ can be written as a $\mathbb{Q}$-linear combination of the v_i's. This implies that an integer scaling of the $\mathbb{Z}$-basis of $\mathfrak{a}_1$ should lie in $\mathfrak{a}_2$. Therefore,

$$\mathbb{C}^\Phi/\Phi(\mathfrak{a}_1) \xrightarrow{\cong} \mathbb{C}^\Phi/\Phi(a\mathfrak{a}_1) \twoheadrightarrow \mathbb{C}^\Phi/\Phi(\mathfrak{a}_2)$$

where the isomorphism is induced by multiplication by a, and the second map is the obvious quotient map because $a\mathfrak{a}_1 \subset \mathfrak{a}_2$ implies $\Phi(a\mathfrak{a}_1) \supset \Phi(\mathfrak{a}_2)$.

So replacing $\mathfrak{a}_1$ by $a\mathfrak{a}_1$, we can assume that $\mathfrak{a}_1 \subset \mathfrak{a}_2$. Now note that the $\mathfrak{a}_i$'s are ideals of $\mathcal{O}_E$ in E. They are invertible since $\mathcal{O}_E$ is a Dedekind Ring. Look at any element α of the $\mathcal{O}_E$-ideal $\mathfrak{a}_1\mathfrak{a}_2^{-1}$, since it is an element of $\mathcal{O}_E$, we get an endomorphism $A_1 \to A_1$, but also $\alpha\mathfrak{a}_2 \subset \mathfrak{a}_1$, therefore the map α factors through A_2. One can see that

$$\Phi(\mathfrak{a}_2)/\Phi(\mathfrak{a}_1) = \bigcap_{\alpha \in \mathfrak{a}_1\mathfrak{a}_2^{-1}} \ker(\alpha)$$

We have already shown that the left side is included in the right. For the other direction, note that if $z \in E$ is such that $\Phi(z) \in \bigcap_{\alpha \in \mathfrak{a}_1\mathfrak{a}_2^{-1}} \ker(\alpha)$, then $\alpha z \in \mathfrak{a}_1$ for all $\alpha \in \mathfrak{a}_1\mathfrak{a}_2^{-1}$. Therefore, $\mathfrak{a}_1\mathfrak{a}_2^{-1} z \in \mathfrak{a}_1$, which means that $z \in \mathfrak{a}_2$. This shows that $\Phi(z) \in \Phi(\mathfrak{a}_2)/\Phi(\mathfrak{a}_1)$ as desired.

Essentially what we did here is show that the natural quotient map

$$\lambda : \mathbb{C}^\Phi/\Phi(\mathfrak{a}_1) \to \mathbb{C}^\Phi/\Phi(\mathfrak{a}_2)$$

has the property that any endomorphism $\alpha \in \mathfrak{a}_1\mathfrak{a}_2^{-1} \subset \mathcal{O}_E = \mathrm{End}(A_1) \cap E$ of A_1 factors through λ. This motivates the following definitions:

Definition 3.6. *Let E be an étale algebra over $\mathbb{Q}$ and let R be an order inside E. Then a* ***lattice ideal*** *$\mathfrak{a}$ of R is an ideal of R that forms a lattice inside E. In other words, it is an ideal that is free as a $\mathbb{Z}$-module of rank the same as the dimension of E over $\mathbb{Q}$.*

We have the following characterisation of a lattice ideal:

Proposition 3.7 (Milne)**.** *Let A be an abelian variety over k with CM by E, and let $R := \mathrm{End}(A) \cap E$. We have that an ideal $\mathfrak{a}$ of R contains an isogeny iff it is a lattice ideal.*

Proof. Note that an element of R is an isogeny iff it is not a zero divisor. An element of E is not a zero divisor iff all its components are non zero (recall $E = \prod_{i=1}^{n} E_i$ is a product of fields).

Now an ideal $\mathfrak{a} \subset R$ contains an isogeny

$$
\begin{aligned}
&\iff \text{it contains } a \in \mathfrak{a} \text{ not a zero divisor} \\
&\iff \text{it contains } a \in \mathfrak{a} \text{ invertible in } E \\
&\iff a = (a_1, \ldots, a_n),\ a_i \neq 0, \text{ is in } \mathfrak{a}. \\
&\implies (a_1, 0, \ldots, 0), (0, a_2, 0, \ldots, 0), \ldots \text{ are in } \mathfrak{a}. \\
&\implies (a_1 R, 0, \ldots, 0), (0, a_2 R, 0, \ldots, 0), \ldots \text{ are in } \mathfrak{a}. \\
&\implies \mathfrak{a} \otimes_{\mathbb{Z}} \mathbb{Q} = E \\
&\implies \mathfrak{a} \text{ is a lattice in } E
\end{aligned}
$$

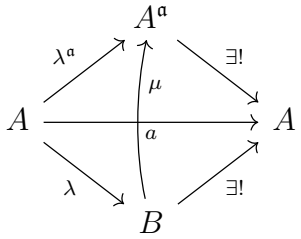
Conversely if $\mathfrak{a}$ is a lattice in E, then it contains some $a = (a_1, \ldots, a_n) \in \mathfrak{a}$ such that $a_i \neq 0$ for all i, and hence it contains an isogeny. □

Remark 3.8. *Note that in our discussion of the complex case above, the ideal $\mathfrak{a}_1 \mathfrak{a}_2^{-1}$ was a lattice ideal.*

Definition 3.9. *Let A be an abelian variety with CM by E over a field k, and let $R := \mathrm{End}(A) \cap E$. Then, for any lattice ideal $\mathfrak{a} \subset R$, an isogeny*

$$\lambda^{\mathfrak{a}} : A \to A^{\mathfrak{a}}$$

is called the ***$\mathfrak{a}$-multiplication****, if any endomorphism $a : A \to A$ with $a \in \mathfrak{a}$, factors uniquely through $\lambda^{\mathfrak{a}}$ and $\lambda^{\mathfrak{a}}$ is the universal object satisfying this property. In other words, for any other isogeny $\lambda : A \to B$ with the same property, there exists a map $\mu : B \to A^{\mathfrak{a}}$ such that $\mu \circ \lambda = \lambda^{\mathfrak{a}}$.*



Remark 3.10. *1. Note that the surjectivity of λ forces μ to be unique.*

2. *The universal property of $\mathfrak{a}$-multiplication shows that an $\mathfrak{a}$-multiplication, if it exists, is unique upto unique isomorphism.*

3. *Let $a \in \mathfrak{a}$ be an isogeny, which exists because $\mathfrak{a}$ is a lattice ideal and we have 3.7, then $\lambda^{\mathfrak{a}}$ is an isogeny since a factors through it.*

Example 3.11. *In our discussion at the beginning of this section, the natural quotient map* $\lambda : \mathbb{C}^\Phi/\Phi(\mathfrak{a}_1) \to \mathbb{C}^\Phi/\Phi(\mathfrak{a}_2)$ *is an* $\mathfrak{a}_1\mathfrak{a}_2^{-1}$ *multiplication. It can be seen that it satisfies the universal property because its kernel is exactly* $\Phi(\mathfrak{a}_2)/\Phi(\mathfrak{a}_1) = \bigcap_{\alpha\in\mathfrak{a}_1\mathfrak{a}_2^{-1}}(\alpha)$.

Proposition 3.12 (Milne)**.** *Given a CM abelian variety* A *with CM by* E *over a field* k, $\mathfrak{a}$*-multiplications exist for any lattice ideal* $\mathfrak{a}$ *of* R*. They can be constructed so that* $A^\mathfrak{a}$ *and* λ^a *are both defined over* k*, the field of definition of* A*.*

Proof. Let $a_1, \dots, a_m$ be generators for $\mathfrak{a}$. Then we define the map:

$$\begin{aligned} A &\xrightarrow{\varphi} A^m \\ x &\mapsto (a_1x, \dots, a_mx) \end{aligned}$$

Now we define $A^\mathfrak{a} := \mathrm{im}(\varphi)$. Note that the map φ is defined over k, since each a_i is an endomorphism of A defined over k. Therefore, $A^\mathfrak{a}$ is also defined over k. Now, for any $a = \sum_{i=1}^m r_ia_i \in \mathfrak{a}$, with $r_i \in R$, we have

$$\begin{gathered} A \xrightarrow{\varphi} A^m \to A \\ x \mapsto (xa_1, xa_2, \dots, xa_m) \mapsto \sum_{i=1}^m r_ia_ix \end{gathered}$$

This shows that any $a \in \mathfrak{a}$ factors through $A^\mathfrak{a}$. Now let λ' be a quotient map $A \to A'$ s.t. any $a \in \mathfrak{a}$ factors through it. So there exists $\alpha_i : A' \to A$ such that $a_i = \alpha_i \circ \lambda'$ for all i. This gives the composite map

$$A \xrightarrow{\lambda'} A' \xrightarrow{\alpha=\begin{pmatrix}\alpha_1\\ \vdots \\ \alpha_m\end{pmatrix}} A^m$$

which is just given by

$$x \mapsto (a_1x, \dots, a_mx)$$

which means $\alpha \circ \lambda' = \lambda^\mathfrak{a}$.

□

We can also compose an $\mathfrak{a}$-multiplication with an $\mathfrak{a}'$-multiplication to get an $\mathfrak{a}\mathfrak{a}'$-multiplication.

Proposition 3.13 (Milne)**.** *Let* $\lambda : A \to B$ *and* $\lambda' : B \to C$ *be* $\mathfrak{a}$ *and* $\mathfrak{a}'$*-multiplications. Then the composition* $\lambda' \circ \lambda$ *is an* $\mathfrak{a}\mathfrak{a}'$*-multiplication.*

Proof. Let $\mathfrak{a}$ have generators $a_1, \dots, a_m$ and $\mathfrak{a}'$ have generators $a'_1, \dots, a'_n$. Then $\mathfrak{a}\mathfrak{a}'$ has generators $\{a_ia'_j\}_{i,j}$. Then we have the following commutative diagram:

$$\begin{array}{ccccc} & & B & \xrightarrow{(a'_1,\dots,a'_n)} & B^n \\ & & \downarrow & & \downarrow \\ A & \xrightarrow{(a_1,\dots,a_m)} & A^m & \xrightarrow{T} & A^{mn} \\ & \searrow^{(a_ia'_j)} & & & \end{array}$$

where T is the matrix given by

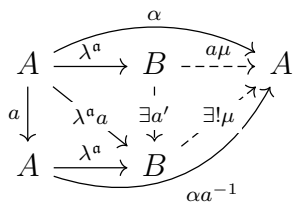
$$T := \begin{pmatrix} a'_1 I_m \\ \vdots \\ a'_n I_m \end{pmatrix}$$

and I_m is the $m \times m$ identity matrix. The commutativity of this diagram allows us to identify C, that is the image of B in B^n, with the image of A in A^{mn}. Hence, $A \to C$ is indeed an $\mathfrak{a}\mathfrak{a}'$-multiplication. □

Proposition 3.14 (Milne)**.** *Given a CM abelian variety A with CM by E over a field k, we have that $A^{\mathfrak{a}}$ is an abelian variety with CM by E over k, and the $\mathfrak{a}$-multiplication map is an E-isogeny. In addition, $R \subset \mathrm{End}(A^{\mathfrak{a}}) \cap E$.*

Proof. We already know that A and $A^{\mathfrak{a}}$ are isogenous, therefore they have the same dimension and their endomorphism algebras $\mathrm{End}^0(A)$ and $\mathrm{End}^0(A^{\mathfrak{a}})$ are also isomorphic and have the same dimensions over $\mathbb{Q}$.

Let $a \in R$ be any element and take an arbitrary $\alpha \in a\mathfrak{a}$. Then we have the following commutative diagram:



Here, the map $\mu : B \to A$ exists because of $a\lambda^{\mathfrak{a}}$ being an $a\mathfrak{a}$-multiplication. Therefore we have

$$\alpha = \mu\lambda^{\mathfrak{a}} a$$

Now note that both $a, \alpha \in E$, therefore they commute and we have

$$\begin{aligned} a^{-1}\alpha &= a^{-1}\mu\lambda^{\mathfrak{a}} a \\ &= \mu\lambda^{\mathfrak{a}} a a^{-1} \\ &= \mu\lambda^{\mathfrak{a}} \\ \implies \alpha &= a\mu\lambda^{\mathfrak{a}} \end{aligned}$$

Thus, α factors through $\lambda^{\mathfrak{a}}$. Therefore, by the universal property of $a\lambda^{\mathfrak{a}}$ being an $a\mathfrak{a}$-multiplication, we have a map $a' : B \to B$ making the diagram commute. It is easy to see that this a' is unique, thus we get an injective map $R \hookrightarrow \mathrm{End}(B)$. This extends to a map $E \hookrightarrow \mathrm{End}^0(B)$ via tensoring with $\mathbb{Q}$. Therefore, B has CM by E over k, and $R \subset \mathrm{End}(B) \cap E$. Now one notes that, through the definition of the inclusion $E \hookrightarrow \mathrm{End}^0(B)$, $\lambda^{\mathfrak{a}}$ is an E-isogeny.

□

We can analyse the degrees of $\mathfrak{a}$-multiplications quite easily in the case of principal CM abelian varieties. To recall the definition of degrees:

Definition 3.15. *Let $\alpha : A \to B$ be an isogeny between abelian varieties (not necessarily with CM), then we have a map of fields $\alpha^* : k(B) \to k(A)$. We define the **degree** of the isogeny α to be*

$$\deg(\alpha) := [k(A) : \alpha^* k(B)]$$

Lemma 3.16 (Milne)**.** *For a CM abelian variety A (not necessarily principal), with CM by E over a field k, we have*

$$\deg(\lambda^{(a)}) = N(a)$$

for any element $a \in R := \mathrm{End}(A) \cap E$ that is not a zero divisor. Here $N(a) := |R/(a)|$.

Proof. Refer to [3], Proposition 7.16. □

Lemma 3.17 (Milne)**.** *For a principal CM abelian variety A, with CM by E over a field k, we have*

$$\deg(\lambda^{\mathfrak{a}}) = N(\mathfrak{a})$$

for any ideal $\mathfrak{a} \subset \mathcal{O}_E = \mathrm{End}(A) \cap E$. Here $N(\mathfrak{a}) := |\mathcal{O}_E/\mathfrak{a}|$.

Proof. Since $\mathcal{O}_E$ is a Dedekind ring, we have $\mathfrak{a} = \prod_{i=1}^{n} \mathfrak{p}_i^{m_i}$ for prime ideals $\mathfrak{p}_i$ and $m_i \in \mathbb{N}$. By the Chinese Remainder theorem, there exists an $a \in \mathcal{O}_E$ such that

$$\begin{aligned}
&a = 0 \pmod{\mathfrak{p}_i^{m_i}} \ \forall\ i = 1, \ldots, n \\
&a = 1 \pmod{\mathfrak{p}_i^{m_i+1}} \ \forall\ i = 1, \ldots, n \\
&a = 1 \pmod{\mathfrak{q}} \ \forall\ \mathfrak{q} \text{ such that } N(\mathfrak{q}) = N(\mathfrak{p}_i) \text{ for some } i = 1, \ldots, n \text{ and } \mathfrak{q} \neq \mathfrak{p}_\mathfrak{i}\ \forall\ i
\end{aligned}$$

Therefore, $a \in \mathfrak{a}$ and we have

$$(a) = \mathfrak{a}\mathfrak{b}$$

for some $\mathfrak{b} \subset \mathcal{O}_E$. Note here that $\mathfrak{b}$ is coprime to $\mathfrak{a}$, in particular $N(\mathfrak{a})$ is coprime to $N(\mathfrak{b})$. Then by 3.13, we see that

$$\deg(\lambda^{(a)}) = \deg(\lambda^{\mathfrak{a}}) \deg(\lambda^{\mathfrak{b}})$$

Claim (Milne)**.** *For a prime $\ell \in \mathbb{Z}$,*

$$\ell \mid \deg(\lambda^{\mathfrak{a}}) \implies \ell \mid N(\mathfrak{a})$$

Proof. If a prime $\ell \in \mathbb{Z}$ doesn't divide $N(\mathfrak{a})$, then $\mathcal{O}_E = \mathfrak{a} + (\ell)$, so by the Chinese Remainder theorem, we can find an element $b \in \mathcal{O}_E$ such that

$$b = 1 \pmod{\ell},\ b = 0 \pmod{\mathfrak{a}}$$

This means that ℓ doesn't divide $N(b)$. We can write

$$(b) = \mathfrak{a}\mathfrak{c}$$

for some ideal $\mathfrak{c}$. Therefore, again by 3.13 and 3.16

$$N(b) = \deg(\lambda^{(b)}) = \deg(\lambda^{\mathfrak{a}}) \deg(\lambda^{\mathfrak{c}})$$

which implies that $\ell \nmid \deg(\lambda^{\mathfrak{a}})$. □

Now since $N(\mathfrak{a})$ and $N(\mathfrak{b})$ are coprime, we get that $\deg(\lambda^{\mathfrak{a}})$ and $\deg(\lambda^{\mathfrak{b}})$ are coprime too. This along with the two equations,

$$\deg(\lambda^{(a)}) = \deg(\lambda^{\mathfrak{a}}) \deg(\lambda^{\mathfrak{b}})$$
$$\deg(\lambda^{(a)}) = N(\mathfrak{a})N(\mathfrak{b})$$

gives us that

$$\deg(\lambda^{\mathfrak{a}}) = N(\mathfrak{a})$$

□

We now have a proposition that allows us to compare and relate different $\mathfrak{a}$-multiplications for principal CM abelian varieties.

Proposition 3.18 (Milne)**.** *Let A be an abelian variety with CM by E over a field k, and assume that $\mathrm{End}(A) \cap E = \mathcal{O}_E$. Let $\lambda^{\mathfrak{a}} : A \to A^{\mathfrak{a}}$ and $\lambda^{\mathfrak{a}'} : A \to A^{\mathfrak{a}'}$ be $\mathfrak{a}$ and $\mathfrak{a}'$-multiplications. Then there exists an E-isogeny $\alpha : A^{\mathfrak{a}} \to A^{\mathfrak{a}'}$ such that $\alpha \circ \lambda^{\mathfrak{a}} = \lambda^{\mathfrak{a}'}$ if and only if $\mathfrak{a} \supset \mathfrak{a}'$.*

Proof. If $\mathfrak{a} \supset \mathfrak{a}'$, then any $a \in \mathfrak{a}'$ factors through $\lambda^{\mathfrak{a}}$. The universality of $\lambda^{\mathfrak{a}'}$ implies that there exists an $\alpha : A^{\mathfrak{a}} \to A^{\mathfrak{a}'}$ such that $\alpha \circ \lambda^{\mathfrak{a}} = \lambda^{\mathfrak{a}'}$. Note that this is an E-isogeny because both $\lambda^{\mathfrak{a}}$ and $\lambda^{\mathfrak{a}'}$ are. Indeed, for any $x \in A^{\mathfrak{a}}$, we have an $x_0 \in A$ such that $x = \lambda^{\mathfrak{a}}(x_0)$, therefore,

$$\alpha(\lambda^{\mathfrak{a}}(x_0)) = \lambda^{\mathfrak{a}'}(x_0)$$

For any element $e \in R$, we get

$$\alpha(\lambda^{\mathfrak{a}}(ex_0)) = \lambda^{\mathfrak{a}'}(ex_0)$$

But since both the λ's are E-isogenies, we have

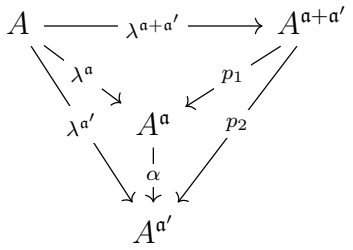
$$\alpha(e\lambda^{\mathfrak{a}}(x_0)) = e\lambda^{\mathfrak{a}'}(x_0)$$

This implies

$$\alpha(ex) = e\alpha(\lambda^{\mathfrak{a}}(x_0)) = e\alpha(x)$$

as desired.

Conversely, if there is such an α, then we look at the natural projection maps $p_1 : A^{\mathfrak{a}+\mathfrak{a}'} \to A^{\mathfrak{a}}$ and $p_2 : A^{\mathfrak{a}+\mathfrak{a}'} \to A^{\mathfrak{a}'}$ obtained from our construction in 3.12. Suppose $p_1(x) = 0$ for some $x \in A^{\mathfrak{a}+\mathfrak{a}'}$, then $p_2(x) = \alpha(p_1(x)) = 0$. Indeed, this is true because of the commutative diagram:



But note that if $p_1(x) = p_2(x) = 0$, then $x = 0$ as well. Therefore, p_1 is an injection. But p_1 is an isogeny as well, hence p_1 is an isomorphism. This implies $\mathfrak{a} = \mathfrak{a} + \mathfrak{a}'$ by 3.17, which implies that $\mathfrak{a} \supset \mathfrak{a}'$. □

This gives the immediate corollary:

Corollary 3.18.1 (Milne)**.** *For a principal CM abelian variety A, let $\lambda^{\mathfrak{a}} : A \to A^{\mathfrak{a}}$ and $\lambda^{\mathfrak{a}'} : A \to A^{\mathfrak{a}'}$ be $\mathfrak{a}$ and $\mathfrak{a}'$-multiplications. Then there exists an E-isomorphism $\alpha : A^{\mathfrak{a}} \to A^{\mathfrak{a}'}$ such that $\alpha \circ \lambda^{\mathfrak{a}} = \lambda^{\mathfrak{a}'}$ if and only if $\mathfrak{a} = \mathfrak{a}'$.*

Proof. The only if part is clear by applying 3.18 to α and its inverse.

For the if part, note that via the universal property of $\mathfrak{a}$-multiplications, $A^{\mathfrak{a}}$ is unique upto unique isomorphism. It is easy to check that this unique isomorphism between $A^{\mathfrak{a}}$ and $A^{\mathfrak{a}'}$ is an E-isomorphism. □

We also have

Corollary 3.18.2 (Milne)**.** *For a principal CM abelian variety A, an element $a \in \mathcal{O}_E$ factors through $\lambda^{\mathfrak{a}}$ iff $a \in \mathfrak{a}$.*

Proof. Use the fact that $a : A \to A$ is an (a)-multiplication. Since a factors through $\lambda^{\mathfrak{a}}$, we have an E-isogeny $\alpha : A^{\mathfrak{a}} \to A$ such that $\alpha \circ \lambda^{\mathfrak{a}} = a$, therefore $(a) \subset \mathfrak{a}$. □

The following result says that any E-isogeny between CM abelian varieties defined over a subfield of $\mathbb{C}$ is an $\mathfrak{a}$-multiplication.

Proposition 3.19 (Milne)**.** *Let A be an abelian variety with CM by E over $k \subset \mathbb{C}$, and assume $\mathcal{O}_E = \mathrm{End}(A) \cap E$. Let $A \xrightarrow{\lambda} B$ be an E-isogeny, then*

1. $\mathrm{Hom}_E(B, A) \to \mathfrak{a} : \mu \mapsto \mu \circ \lambda$ *is an isomorphism, for some $\mathfrak{a}$ an ideal of $\mathcal{O}_E$.*
2. *λ is an $\mathfrak{a}$-multiplication.*

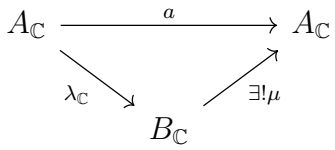
Proof. We have the map

$$\mathrm{Hom}_E(B, A) \to \mathrm{End}(A) : \mu \mapsto \mu \circ \lambda$$

and $\mu\lambda$ commutes with everything in $\mathcal{O}_E$, therefore $\mu\lambda \in \mathcal{O}_E$ due to 3.4. Now since $\mathrm{Hom}_E(B, A)$ is an $\mathcal{O}_E$-module, we get that its image in $\mathcal{O}_E$ is an $\mathcal{O}_E$-ideal, call it $\mathfrak{a}$. One must note that the map $\mu \mapsto \mu \circ \lambda$ is injective since λ is an isogeny and so is in particular surjective. This shows that we have the required isomorphism. For the second part we have

$$\varphi \mapsto \varphi \circ \lambda^{\mathfrak{a}} : \mathrm{Hom}_E(A^{\mathfrak{a}}, A) \xrightarrow{\cong} \mathfrak{a}$$

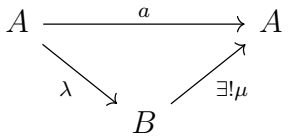
This is an isomorphism because we can use the same logic as in the case of $\mathrm{Hom}_E(B, A)$ to get that the map injects into $\mathcal{O}_E$, and so is an ideal in $\mathcal{O}_E$. Also, we can use the fact that any $a \in \mathcal{O}_E$ factors through $\lambda^{\mathfrak{a}}$ iff $a \in \mathfrak{a}$ (see 3.18.2). So if we just show that λ is a $\mathfrak{b}$-multiplication for some ideal $\mathfrak{b} \subset \mathcal{O}_E$, we can conclude from the first part that $\mathfrak{b} = \mathfrak{a}$, and that λ is an $\mathfrak{a}$-multiplication. Note that if $\lambda : A \to B$ is an isogeny defined over k, then $\lambda_{\mathbb{C}} : A_{\mathbb{C}} \to B_{\mathbb{C}}$ is an isogeny over $\mathbb{C}$, where $X_{\mathbb{C}}$ denotes the base change to $\mathbb{C}$ for $X = A, B, \lambda$. Indeed this is true because a base change preserves surjectivity. But we showed in the beginning of this section that any isogeny between CM abelian varieties over $\mathbb{C}$ is an $\mathfrak{b}$-multiplication for some ideal $\mathfrak{b}$. Therefore we have the commutative diagram



for any $a \in \mathfrak{b}$. But now, for any $\sigma \in \mathrm{Gal}(\mathbb{C}/k)$ and any $y \in B_{\mathbb{C}}$, there exists an $x \in A_{\mathbb{C}}$ such that $\lambda_{\mathbb{C}}(x) = y$ (because $\lambda_{\mathbb{C}}$ is an isogeny). Therefore,

$$\begin{aligned}
\mu(\sigma y) &= \mu(\sigma \lambda_{\mathbb{C}} x) \\
&= \mu(\lambda_{\mathbb{C}} \sigma x) \\
&= a(\sigma x) \\
&= \sigma(ax) \\
&= \sigma(\mu \lambda_{\mathbb{C}} x) \\
&= \sigma(\mu y)
\end{aligned}$$

Here we used the fact that λ and a are defined over k, and showed that μ is defined over k. Therefore we get the diagram:



But now, if there is a $\lambda' : A \to B'$ through which every $a \in \mathfrak{b}$ factors, Then we know by the universal property of $\mathfrak{b}$-multiplications, that there exists a $\varphi : B_{\mathbb{C}} \to B'_{\mathbb{C}}$ such that $\lambda'_{\mathbb{C}} = \varphi \circ \lambda_{\mathbb{C}}$. Now one can show like we did above, that φ is defined over k, and so we get $\lambda' = \varphi \circ \lambda$. This shows that λ is a $\mathfrak{b}$-multiplication. □

Lemma 3.20. *Let A, B be two abelian varieties with CM by E over $k \subset \mathbb{C}$. Let $\alpha, \beta : A \to B$ be two E-isogenies. Then there exists $e \in E^\times$ such that $\alpha \circ e = \beta$, where e is seen as an element of* $\mathrm{End}^0(A)$.

Proof. Let $e_1 \in H_1(A, \mathbb{Q})$ and $e_2 \in H_1(B, \mathbb{Q})$ is the image of e_1 under the isogeny α. Let $\mathfrak{a} \subset E$ be the lattice such that $e_1\mathfrak{a} = H_1(A, \mathbb{Z})$ and similarly, let $\mathfrak{b}$ be the same for B, i.e., $e_2\mathfrak{b} = H_1(B, \mathbb{Z})$. Any E-isogeny $A \to B$ is determined by where e_1 goes. Since $H_1(A, \mathbb{Q}) \to H_1(B, \mathbb{Q})$ is an E-linear map and this induces a map $\mathbb{C}^\Phi \to \mathbb{C}^\Phi$ inducing a map on $A \to B$. Now if we let $e_1 \mapsto e_2'$ be the image of e_1 under β, this defines β completely. So, $\beta(x) = \alpha(x) \cdot \frac{e_2'}{e_2}$ for all $x \in H_1(A, \mathbb{Q})$. But α is E-linear, so we get

$$\beta(x) = \alpha\left(\frac{e_2'}{e_2} \cdot x\right)$$

So $e := e_2'/e_2 \in E$ is the promised element. One must note that both e_2 and e_2' are non zero-divisors in E, precisely because their E-span is all of E. Therefore, they are actually invertible and this justifies the above steps. Also, this shows that $e \in E^\times$. □

Another important proposition to be used in the next chapter is about lifting isogenies on good reductions to $\mathfrak{a}$-multiplications on the global varieties.

Proposition 3.21 (Milne)**.** *Let A, B be principal CM abelian varieties with CM by E over a field k. Let A_0, B_0 denote their good reductions at a prime $\mathfrak{p}$ of $\mathcal{O}_k$. If A, B are isogenous and there is an E-isogeny*

$$\lambda_0 : A_0 \to B_0$$

then there is a lift

$$\lambda : A \to B$$

of λ_0, that is an $\mathfrak{a}$-multiplication for some lattice ideal $\mathfrak{a} \in \mathcal{O}_E$, possibly over an extension of k.

Proof. Proposition 7.42 of [3]. □

Chapter 4

Shimura-Taniyama formula

In this chapter, we will state and prove the Shimura-Taniyama formula, which is essentially the local picture version of Shimura's main theorem of complex multiplication. We closely follow Milne's exposition, supplying some more details where necessary, hoping to make this exposition more self contained.

In the next chapter, we will use this local formula to prove the main theorem.

4.1 Tangent Spaces of Reductions

We will take a short detour into a bit of scheme theory to formalise some parts of Milne's proof of the local formula. In particular, the result we wanna make sense of and prove is the following:

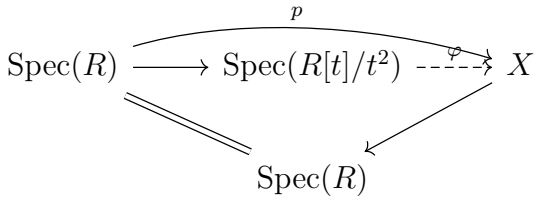
Proposition 4.1. *Let $\mathcal{A}$ be an abelian scheme over a DVR R with field of fractions k and residue field k_0. Let A be the generic fiber of $\mathcal{A}$, and let A_0 be the special fiber, so they are abelian varieties over k and k_0 respectively.*

Let $0 \in \mathcal{A}(R)$ be the R-point on $\mathcal{A}$ corresponding to the zero element and we denote its images in $A(k)$ and in $A_0(k_0)$ as 0 as well. Then $T_0(\mathcal{A})$, the tangent space of $\mathcal{A}$ at 0, is a free R-module of rank $n = \dim_R A$ and we have isomorphisms

$$T_0(A) \cong T_0(\mathcal{A}) \otimes_R k; \quad T_0(A_0) \cong T_0(\mathcal{A}) \otimes_R k_0$$

We recall the definition of the tangent space first.

Definition 4.2. *Let X be a scheme over a ring R, then for an R-point $p \in X(R)$, we define the tangent space $T_p(X)$ at p as the R-module of maps φ that make the following diagram commute:*



One can show that this set of dotted maps φ that complete the diagram above form an R-module, following [9, Tag 0B28].

Lemma 4.3. *Let R be a DVR and X a smooth scheme of relative dimension n over it. Then for any point $p \in X(R)$, $T_p(X)$ is a free R-module of rank equal to that of the sheaf of differentials at p as an $\mathcal{O}_{X,p}$-module. In particular,*

$$T_p(X) \cong \mathrm{Hom}_{\mathcal{O}_{X,p}}(\Omega_{X/R,p}, R) \cong R^n$$

where $n = \mathrm{rank}_{\mathcal{O}_{X,p}}(\Omega_{X/R,p}) = \dim_R(X)$.

Proof. From [9, Lemma 02G1], we know that for a smooth morphism $X \to \mathrm{Spec}(R)$, one has

$$\mathrm{rank}_{\mathcal{O}_{X,p}}(\Omega_{X/R,p}) = \dim_R(X)$$

where $\dim_R X$ is the relative dimension of X over $\mathrm{Spec}(R)$. We know that R has an $\mathcal{O}_{X,p}$-module structure coming from the map on stalks induced by $p : \mathrm{Spec}(R) \to X$.

Now note that any element $\theta \in T_p(X)$ can be considered as a local ring map $\theta : \mathcal{O}_{X,p} \to R[t]/(t^2)$ that is also an R-algebra map. It can be written as $\theta(a) = c(a) + D(a)t$ where $c : \mathcal{O}_{X,p} \to R$ is the map coming from the $p : \mathrm{Spec}(R) \to X$. One sees that D is an R-derivation from $\mathcal{O}_{X,p}$ to the $\mathcal{O}_{X,p}$-module R. Indeed, note that c is an R-algebra map, hence is R-linear, which along with R-linearity of θ gives R-linearity of D. Also,

$$\theta(ab) = \theta(a)\theta(b) \implies c(ab) + D(ab)t = c(a)c(b) + t(c(a)D(b) + c(b)D(a))$$

which gives $D(ab) = c(a)D(b) + c(b)D(a)$.

Conversely, given an R-derivation $D : \mathcal{O}_{X,p} \to R$, one notes that $\theta := c + D \cdot t$ is an element of $T_p(X)$.

So any element θ is in 1-1 correspondence with an R-derivation $\mathcal{O}_{X,p} \to R$, which is in 1-1 correspondence with a $\mathcal{O}_{X,p}$-linear map $\Omega_{X/R,p} \to R$. This finishes the proof. □

Lemma 4.4. *With the same notation as 4.3, we have for any commutative ring map $R \to S$,*

$$T_{p_S}(X_S) \cong T_p(X) \otimes_R S$$

where $X_S := X \times_{\mathrm{Spec(R)}} \mathrm{Spec}(S)$, and p_S is the unique S-point on X_S induced by $p \in X(R)$.

Proof. From 4.3, we see that

$$R^n = T_p(X) = \mathrm{Hom}_{\mathcal{O}_{X,p}}(\Omega_{X/R,p}, R)$$

and since X_S is smooth over S, we get that

$$S^n = T_{p_S}(X_S) = \mathrm{Hom}_{\mathcal{O}_{X_S,p_S}}(\Omega_{X_S/S,p_S}, S)$$

here we used the fact that relative dimension is preserved under base change ([9, Lemma 02NK]) and the result follows. □

Now we are ready to prove 4.1.

proof of 4.1. Note that the abelian scheme $\mathcal{A}$ is a smooth scheme over the domain R, so we can apply 4.3 and 4.4, the proposition follows immediately.

□

Now let A be an abelian variety with CM by E over a field k with char 0, which can be embedded into $\mathbb{C}$ and which contains all conjugates of E.

Note that being embeddable into $\mathbb{C}$ is not too bad of an assumption, because any field extension of $\mathbb{Q}$ with transcendence degree $\leq 2^{\aleph_0}$ has this property (since the transcendence degree of $\mathbb{C}$ over $\mathbb{Q}$ is $2^{\aleph_0}$). Since we will be concerned only with algebraic number fields later, we can safely make this assumption for now.

Remark 4.5. *As noted in 1.19, the tangent space of a CM abelian variety turns out to be the trace module defined in 1.18. Indeed, one notes that $T := \mathrm{T}_0(A)$ is a finitely generated module over $E \otimes_{\mathbb{Q}} k$ since it is a k-vector space with an action of E via complex multiplication. More precisely, any element $a \in \mathrm{End}(A) \cap E$ induces a map on the tangent space of A, and this can be extended to an action of any element of E via scaling by rationals.*

Note that $E \otimes_{\mathbb{Q}} k = \prod_{E \to k} k$ is a semi-simple k-algebra. Its simple modules are denoted by k_φ which are just copies of k with the action of E given by $\varphi : E \to k$. Therefore the $E \otimes_{\mathbb{Q}} k$-module $T_0(A)$ is a product $\prod_{\varphi : E \to k} k_\varphi$ where φ runs over a subset (possibly a multiset) of embeddings of E into k. We also know from section 2.5 that

$$T_0(A_{\mathbb{C}}) = \prod_{\varphi \in \Phi'} \mathbb{C}_\varphi$$

where $\Phi' \subset \mathrm{Hom}(E, \mathbb{C})$ such that $\Phi' \sqcup \iota\Phi' = \mathrm{Hom}(E, \mathbb{C})$. But $T_0(A_{\mathbb{C}}) = T_0(A) \otimes_k \mathbb{C}$ by 4.4. Therefore,

$$T_0(A) = \prod_{\varphi \in \Phi} k_\varphi$$

where $\Phi \subset \mathrm{Hom}(E, k)$ such that $\Phi \sqcup \iota\Phi = \mathrm{Hom}(E, k)$. This also allows us to assign a CM-type to the CM abelian variety A.

4.2 Some technical results

We state some technical results needed for our proof of the local formula, we omit proofs and refer the reader to Milne for the proofs.

Proposition 4.6. *(Milne) Let $\alpha : A \to B$ be an isogeny of abelian varieties over an algebraically closed field k of characteristic $p > 0$. Assume that $\alpha^*(k(B)) \subset k(A)^q$ for some power $q = p^\ell$ of p, and let k be the dimension of the kernel of $T_0(\alpha) : T_0(A) \to T_0(B)$. Then*

$$\deg(\alpha) \leq q^k.$$

Proof. Proposition 7.17 of [3]. □

Proposition 4.7. *Let A_0 be an abelian variety over a finite field $\mathbb{F}_q$, and let* Frob *denote the frobenius endomorphism, i.e., the q-th power endomorphism on A_0. Then the degree of* Frob *is given by*

$$\deg(\mathrm{Frob}) = q^n$$

where $n = \dim A_0$. In addition we also have

$$\mathrm{Frob}^\dagger \circ \mathrm{Frob} = [q]$$

where $\mathrm{Frob}^\dagger$ *is the image of* Frob *under the Rosati involution (under some chosen polarisation) and $[q]$ is the isogeny sending $P \mapsto qP$ for all points $P \in A_0$.*

Proof. Lemma 1.2 in § II of [2] shows that $\text{Frob}^\dagger \circ \text{Frob} = [q]$. Now let $k(A_0)$ be the function field of A_0. This is a field extension of $\mathbb{F}_q$ with transcendence degree n over it. So it has the form $\mathbb{F}_{q^m}(t_1, \ldots, t_n)$ for independent variables $t_1, \ldots, t_n$ and some $m \in \mathbb{N}$. Then we have

$$\deg(\text{Frob}) = [k(A_0) : \text{Frob}^*(k(A_0))] = [\mathbb{F}_{q^m}(t_1, \ldots, t_n) : \mathbb{F}_{q^m}(t_1^q, \ldots, t_n^q)] = q^n$$

□

4.3 The formula

We begin by diving straight into the statement.

Theorem 4.8 (Milne)**.** *Let A be an abelian variety with CM by an étale $\mathbb{Q}$-algebra E and CM-type Φ over a field k that is a finite extension of $\mathbb{Q}_p$ for some prime p. Suppose A has good reduction at the prime $\mathfrak{p} \subset \mathcal{O}_k$ above $(p) \subset \mathbb{Z}_p$. Assume p is unramified in E and $\text{End}(A) \cap E = \mathcal{O}_E$. Then,*

(a) there exists an element $\pi \in \mathcal{O}_E$ inducing Frob_q on the reduction of A, where $q = |\mathcal{O}_k/\mathfrak{p}|$.

(b) Now suppose k contains all conjugates of E, then

$$(\pi) = \prod_{\varphi \in \Phi} \varphi^{-1}(\text{Nm}_{k/\varphi E}\mathfrak{p}) = N_{k,\Phi}(\mathfrak{p})$$

This has the following important corollary.

Corollary 4.8.1 (Milne)**.** *Let (E, Φ) be a CM-type and A be a principal CM abelian variety of CM-type (E, Φ) over a number field k which is a Galois extension of $\mathbb{Q}$. Suppose A has good reduction at $\mathfrak{P} \subset \mathcal{O}_k$. Let p be the rational prime below $\mathfrak{P}$, and $\mathfrak{p}$ be the prime in E^* below $\mathfrak{P}$. Suppose p is unramified in E, and $\mathfrak{P}$ is unramified over E^*. Let $\sigma := (\mathfrak{P}, k/E^*)$ be the frobenius element in $\text{Gal}(k/\mathbb{Q})$ associated to $\mathfrak{P}$. Then,*

1. *there exists an $\mathfrak{a}$-multiplication $\alpha : A \to \sigma A$ such that its reduction at $\mathfrak{P}$ to A_0 is the q-power frobenius, where $q = |\mathcal{O}_{E^*}/\mathfrak{p}|$.*
2. $\mathfrak{a} = N_\Phi(\mathfrak{p})$

proof of 4.8.1. A and σA are of the same CM-type, they are isogenous. So any $\mathcal{O}_E$-isogeny between their reductions lifts to an $\mathfrak{a}$-multiplication over a finite extension of k between A and σA (due to 3.21) for some lattice ideal $\mathfrak{a}$. Since the q-power frobenius is an $\mathcal{O}_E$-isogeny, we get the first part.

For the second part, note that

$$\sigma^{f-1}\alpha \circ \cdots \circ \sigma\alpha \circ \alpha = \pi$$

for some $\pi \in \mathcal{O}_E$. Indeed, the composition on the left reduces to the q^f-power frobenius map on A_0, where $f = [\mathcal{O}_k/\mathfrak{P} : \mathcal{O}_{E^*}/\mathfrak{p}]$ and we know from 4.8, that π represents this element of $\text{End}(A)$. Now,

$$\mathfrak{a}^f = N_{k,\Phi}(\mathfrak{P}) = N_\Phi(\text{Nm}_{k/E^*}\mathfrak{P}) = N_\Phi(\mathfrak{p}^f) = N_\Phi(\mathfrak{p})^f$$

Therefore,

$$\mathfrak{a} = N_\Phi(\mathfrak{p})$$

□

We use the following lemma, whose proof has been adapted from §13.2 of [6].

Lemma 4.9 (Shimura)**.** *Consider the notations of 4.8. Let $\mathcal{A}$ be the abelian scheme over $\mathcal{O}_k$ such that its generic fiber is A. Then we can find an $\mathcal{O}_k$-basis $\{e_\varphi \mid \varphi \in \Phi\}$ of $T_0(\mathcal{A})$ such that $a \cdot e_\varphi = \varphi(a) \cdot e_\varphi$ for all $a \in E$.*

Proof. Let $\{v_\varphi \mid \varphi \in \Phi\}$ be a basis of $T_0(A)$ over k s.t. $a \cdot v_\varphi = \varphi(a)v_\varphi$. We can scale them so that each $v_i \in \mathcal{T} := T_0(\mathcal{A})$ and $\tilde{v}_i \neq 0$ (the reductions of v_i in the quotient $\mathcal{T}/\mathfrak{p}\mathcal{T} = T_0(A_0)$) for all i. Let $\beta_1, \ldots, \beta_{2n}$ be a basis of $\mathcal{O}_E/\mathbb{Z}$. Define

$$A := \begin{bmatrix} \varphi_1(\beta_1) & \ldots & \varphi_n(\beta_1) & \varphi_1\iota_E(\beta_1) & \ldots & \varphi_n\iota_E(\beta_1) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \varphi_1(\beta_{2n}) & \ldots & \varphi_n(\beta_{2n}) & \varphi_1\iota_E(\beta_{2n}) & \ldots & \varphi_n\iota_E(\beta_{2n}) \end{bmatrix}$$

One sees that $\det(A) = \mathrm{disc}_{\mathbb{Q}}(E)$. Now we know that p is unramified in E, this is equivalent to saying that $\mathrm{disc}_{\mathbb{Q}}(E) \neq 0 \pmod{p}$, which is the same as saying $\det(A) \neq 0 \pmod{p}$. Since the determinant is non zero, we see that the first n-columns have rank n. We can rearrange the rows to get that the top left $n \times n$ square has $\det \neq 0 \bmod p$. To say that $v_1, \ldots, v_n$ form a basis for $\mathcal{T}$ over $\mathcal{O}_k$, it is enough to say that $\det(v_i)_{i=1}^n \in \mathcal{O}_k \setminus \mathfrak{p}$, which is the same as saying that $\tilde{v}_i$ forms a basis over k_0 for $T_0(A_0)$. This is exactly what we show.

Let $c_i \in k_0$ for $i = 1, \ldots, n$ such that

$$\sum_{i=1}^{n} c_i \tilde{v}_i = 0$$

Apply β_j to this relation

$$\sum_{i=1}^{n} c_i(\varphi_i(\beta_j)\tilde{v}_i) = 0$$

This means

$$\sum_{i=1}^{n} c_i \tilde{v}_i \begin{bmatrix} \varphi_i(\beta_1) \\ \varphi_i(\beta_2) \\ \vdots \\ \varphi_i(\beta_n) \end{bmatrix} = 0$$

Let $p : T_0(A_0) \to k_0$ be any linear functional. Then applying p to the above equation, we get

$$\sum_{i=1}^{n} c_i p(\tilde{v}_i) \begin{bmatrix} \varphi_i(\beta_1) \\ \varphi_i(\beta_2) \\ \vdots \\ \varphi_i(\beta_n) \end{bmatrix} = 0$$

But the columns of the top left $n \times n$ submatrix of A are linearly independent, therefore we get

$$c_i p(\tilde{v}_i) = 0 \ \forall i, \ \forall p$$

This means in particular that all c_i's are zero and we get that $\tilde{v}_i$'s are linearly independent. □

proof of 4.8. Let A_0 be the reduction of A to $k_0 := \mathcal{O}_K/\mathfrak{p}$. First note that A_0 is again a CM abelian variety with CM by E. This can be seen by considering the Néron model $\mathcal{A}$ of A, which is an abelian scheme over $\mathcal{O}_K$, and whose special fiber is A_0. It is known that $\mathrm{End}A \cong \mathrm{End}\mathcal{A}$. Now, let $\alpha : \mathcal{A} \to \mathcal{A}$ and let α_0 be its reduction to A_0. Therefore, any element of $\mathrm{End}A$ has a reduction to $\mathrm{End}A_0$.

Let A be simple, then $\mathrm{End}^0(A) \to \mathrm{End}^0(A_0)$ is injective. Indeed, note that the map is given by the isomorphism $\mathrm{End}(A) \cong \mathrm{End}(\mathcal{A})$ which allows us to lift endomorphisms of A to those of $\mathcal{A}$, and the fact that any endomorphism on $\mathcal{A}$ induces an endomorphism on A_0. So we have a map of algebras $\mathrm{End}^0(A) \to \mathrm{End}^0(A_0)$ where the domain is a simple algebra. Since the kernel of this map is an ideal of $\mathrm{End}^0(A)$, it can only be 0 or $\mathrm{End}^0(A)$. But one notes that the identity on A lifts to the identity on $\mathcal{A}$ and so descends to the identity on A_0, therefore ruling out the case when the kernel is $\mathrm{End}^0(A)$.

In general, let $A \sim \prod_{i=1}^m A_i^{n_i}$ with A_i simple which implies that

$$A_0 \sim \prod_{i=1}^m (A_i)_0^{n_i}$$

Therefore, $\mathrm{End}^0(A) = \prod_{i=1}^m M_{n_i}(\mathrm{End}^0(A_i))$. Now, any map $A \to A$ if non trivial, maps some A_i factor into another A_i factor non trivially, and so its reduction maps some $(A_i)_0$ into another $(A_i)_0$ non trivially (since $\mathrm{End}^0(A) \to \mathrm{End}^0(A_0)$ is injective when A is simple as we showed before), and so the reduction is non trivial. So we have that the map $\mathrm{End}^0(A) \to \mathrm{End}^0(A_0)$ is injective. Therefore, A_0 also has complex multiplication by E. Now also note that $\mathcal{O}_E = \mathrm{End}(A) \cap E \subset \mathrm{End}(A_0) \cap E$, but $\mathrm{End}(A_0) \cap E$ is an order in E, hence is a subset of $\mathcal{O}_E$, therefore we get that $\mathrm{End}(A_0) \cap E = \mathcal{O}_E$. The Frobenius endomorphism on A_0 commutes with all endomorphisms of A_0 (this can be seen by thinking in terms of polynomials defining A_0 and the Frobenius being the q-th power map), thus it lies in $\mathcal{O}_E$ due to 3.4, we call this element π and this proves part (a).

For part (b), define $T := T_0(A)$ as the tangent space of A at 0, and $\mathcal{T} := T_0(\mathcal{A})$ as the tangent space of $\mathcal{A}$ at 0. Note that $\mathcal{T}$ is a lattice in T such that $\mathcal{T}/\mathfrak{p}\mathcal{T} = T_0(A_0)$ due to 4.1. By 4.9, we get a basis $\{e_\varphi \mid \varphi \in \Phi\}$ of $\mathcal{T}$. We know the element $\pi \in \mathcal{O}_E$ acts via Frob_q on A_0 and so $\pi\bar{\pi} = q$ because of 4.7 and the fact that the Rosati involution acts via conjugation on E. Therefore,

$$(\pi) = \prod_{\nu|p} \mathfrak{p}_\nu^{m_\nu};\ m_\nu \in \mathbb{Z}_{\geq 0}$$

Let h be the class number of E and define $\gamma_\nu \in \mathcal{O}_E$ such that

$$\mathfrak{p}_\nu^{m_\nu h} = (\gamma_\nu)$$

Note that any $\mathfrak{p}_\nu$ will be a pullback of $\mathfrak{p}$ along some φ_i, therefore we define the partition of Φ into

$$\Phi_\nu := \{\varphi \in \Phi \mid \varphi^{-1}(\mathfrak{p}) = \mathfrak{p}_\nu\}$$

and $d_\nu := |\Phi_\nu|$. Now γ_ν acts on $T_0 := T_0(A_0)$ via multiplication by $\varphi(\gamma_\nu)$ on each $\tilde{e}_\varphi$. Therefore we have

$$\ker(\gamma_\nu|_{T_0}) = \langle e_\varphi \mid \varphi(\gamma_\nu) \in \mathfrak{p}\rangle = \langle e_\varphi \mid \varphi \in \Phi_\nu\rangle$$

Note that $\pi^h : A_0 \to A_0$ factors through γ_ν and so we get that $\gamma_\nu^* k(A_0) \supset (\pi^h)^* k(A_0) = k(A_0)^{q^h}$, where $k(X)$ is the function field of a variety X.

So by 4.6, we get

$$\deg(A_0 \xrightarrow{\gamma_\nu} A_0) \leq q^{hd_\nu}$$

Now,

$$\begin{aligned}\deg(\gamma_\nu|_{A_0}) &= \mathrm{Nm}_{E/\mathbb{Q}}(\gamma_\nu) \quad \text{due to 3.16} \\ &= \mathrm{Nm}_{E/\mathbb{Q}}(\mathfrak{p}_\nu^{hm_\nu}) \\ &\leq q^{hd_\nu}\end{aligned}$$

This gives $\mathrm{Nm}_{E/\mathbb{Q}}(\mathfrak{p}_\nu^{m_\nu}) \leq q^{d_\nu}$. Taking the product over all ν, we get

$$\mathrm{Nm}(\pi) \leq q^n$$

But $\mathrm{Nm}(\pi) = \deg(\pi : A_0 \to A_0) = q^n$. This means we must have equality everywhere in our calculations above. So we get

$$\begin{aligned}\mathrm{Nm}_{E/\mathbb{Q}}(\mathfrak{p}_\nu^{m_\nu}) &= q^{d_\nu} \\ &= \mathrm{Nm}_{k/\mathbb{Q}}(\mathfrak{p})^{d_\nu} \\ &= \prod_{\varphi \in \Phi_\nu} (\mathrm{Nm}_{k/\mathbb{Q}}(\mathfrak{p})) \\ &= \prod_{\varphi \in \Phi_\nu} \mathrm{Nm}_{E/\mathbb{Q}}(\varphi^{-1}\mathrm{Nm}_{k/\varphi E}(\mathfrak{p})) \\ &= \mathrm{Nm}_{E/\mathbb{Q}}\left(\prod_{\varphi \in \Phi_\nu} \varphi^{-1}\mathrm{Nm}_{k/\varphi E}(\mathfrak{p})\right)\end{aligned}$$

This implies that

$$\mathfrak{p}_\nu^{m_\nu} = \prod_{\varphi \in \Phi_\nu} \varphi^{-1}\mathrm{Nm}_{k/\varphi E}(\mathfrak{p})$$

because for $\varphi \in \Phi_\nu$, $\varphi^{-1}\mathrm{Nm}_{k/\varphi E}(\mathfrak{p})$ is a power of $\mathfrak{p}_\nu$ by the definition of Φ_ν. This gives us

$$(\pi) = \prod_{\varphi \in \Phi} \varphi^{-1}\mathrm{Nm}_{k/\varphi E}(\mathfrak{p}) = N_{k,\Phi}(\mathfrak{p})$$

where the last equality is due to 1.25.1. □

Chapter 5

The Main theorem of CM

In this chapter, we state and prove the main theorem of CM using the Shimura-Taniyama formula proved in the last chapter. As promised, this theorem is essentially the global version of the picture picture described by Shimura-Taniyama, in that it describes how elements of Galois groups of extensions of the reflex field act as $\mathfrak{a}$-multiplications on our CM abelian variety. This will be used in the next chapter to construct class fields of the reflex field.

5.1 Preliminaries

We collect preliminary facts and lemmas that we will need in our proof of the main theorem here.

5.1.1 Algebraic Number theory

Let k be an algebraic number field. Denote by $\mathrm{rec}_k : \mathbb{A}_k^\times \to \mathrm{Gal}(k^{\mathrm{ab}}/k)$ the Artin reciprocity map and by $\mathrm{art}_k : \mathbb{A}_k^\times \to \mathrm{Gal}(k^{\mathrm{ab}}/k)$ defined as $\mathrm{art}_k := \mathrm{rec}_k^{-1}$ (by which we mean the reciprocal of the rec_k map). Note that art_k factors through the finite ideles $\mathbb{A}_{f,k}^\times$ when k is totally imaginary. Also, we know that $\mathrm{art}_k|_{\mathbb{A}_{f,k}^\times}$ is surjective with kernel $= \bar{k}^\times$. We define the cyclotomic character:

Definition 5.1. *Let $\chi : \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \to \hat{\mathbb{Z}}^\times$ be defined such that*

$$\sigma\zeta = \zeta^{\chi(\sigma)}$$

for all roots of unity ζ in $\mathbb{C}$.

By the notation $\zeta^{\chi(\sigma)}$ we mean, $\zeta^{\chi(\sigma)_n}$ where n denotes any positive integer divisible by the order of ζ, and $\chi(\sigma)_n$ is the image of $\chi(\sigma)$ in $(\mathbb{Z}/n\mathbb{Z})^\times$ under the projection map $\hat{\mathbb{Z}}^\times \to (\mathbb{Z}/n\mathbb{Z})^\times$. This is well defined since the projection maps to different n's are compatible.

Lemma 5.2 (Milne)**.** *For any $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$, we have*

$$\mathrm{art}_{\mathbb{Q}}(\chi(\sigma)) = \sigma|_{\mathbb{Q}^{\mathrm{ab}}}$$

Proof. We show that the two sides agree on $\mathbb{Q}(\zeta_n)$ for all n. But we have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) & \longleftarrow & \hat{\mathbb{Z}}^\times \\ \downarrow & & \downarrow \\ \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) & \longleftarrow & (\mathbb{Z}/n\mathbb{Z})^\times \end{array}$$

where the top arrow is $\mathrm{art}_{\mathbb{Q}}$, and the bottom map sends a to $\zeta_n \mapsto \zeta_n^{a^{-1}}$. It is easy to see now that by definition of χ, the two sides agree on $\mathbb{Q}(\zeta_n)$. □

Lemma 5.3 (Milne)**.** *Let E be a CM-field, then for any $s \in \mathbb{A}_{f,E}^{\times}$, we have*

$$\mathrm{Nm}_{E/\mathbb{Q}}(s) \in \chi(\mathrm{art}_E(s)) \cdot \mathbb{Q}_{>0}$$

Proof. This is Lemma 9.5 of [3]. Take $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/E)$ such that $\mathrm{art}_E(s) = \sigma|_{E^{\mathrm{ab}}}$. Then

$$\mathrm{art}_{\mathbb{Q}}(\mathrm{Nm}_{E/\mathbb{Q}}(s)) = \sigma|_{\mathbb{Q}^{\mathrm{ab}}}$$

by CFT. Therefore,

$$\mathrm{art}_{\mathbb{Q}}(\mathrm{Nm}_{E/\mathbb{Q}}(s)) = \mathrm{art}_{\mathbb{Q}}(\chi(\sigma))$$

Now one must note that the kernel of the artin map $\mathrm{art}_{\mathbb{Q}}$ is the closure of $\mathbb{Q}^{\times} \times \mathbb{R}_{>0}$. But we know that

$$\mathbb{A}_{\mathbb{Q}}^{\times} \cong \mathbb{Q}^{\times} \times \mathbb{R}_{>0} \times \prod_p \mathbb{Z}_p^{\times}$$

as topological groups. So the kernel is exactly $\mathbb{Q}^{\times} \times \mathbb{R}_{>0}$. We also know that

$$\mathbb{A}_{f,\mathbb{Q}}^{\times} \cong \mathbb{Q}_{>0} \times \prod_p \mathbb{Z}_p^{\times}$$

where the inclusion $\mathbb{A}_{f,\mathbb{Q}}^{\times} \hookrightarrow \mathbb{A}_{\mathbb{Q}}^{\times}$ is given by

$$(a_{\mathbb{Q}}, a_2, a_3, \dots) \mapsto (a_{\mathbb{Q}}, 1, a_2, a_3, \dots)$$

Therefore, the kernel of $\mathrm{art}_{\mathbb{Q}}|_{\mathbb{A}_f^{\times}}$ is indeed given by $\mathbb{Q}_{>0}$. This means that

$$\mathrm{Nm}_{E/\mathbb{Q}}(s) \in \chi(\mathrm{art}_E(s)) \cdot \mathbb{Q}_{>0}$$

as desired. □

5.1.2 Algebraic Tori

Recall the reflex norm introduced in section 1.4. The functoriality of the reflex norm $N_{k,\Phi}$ allows us to view it as a map of tori:

$$N_{k,\Phi} : T^k \to T^E$$

where $T^X := (\mathbb{G}_m)_{X/\mathbb{Q}}$ is the Weil restriction of the multiplicative group $\mathbb{G}_m$, where X is any finite $\mathbb{Q}$-algebra. Note that $T^X(R) = (X \otimes_{\mathbb{Q}} R)^{\times}$ for any $\mathbb{Q}$-algebra R.

Now we know that

$$N_\Phi(a) \cdot \iota_E N_\Phi(a) = \mathrm{Nm}_{k\otimes_{\mathbb{Q}} R/R}(a), \ \forall\ a \in (k \otimes_{\mathbb{Q}} R)^{\times} \tag{5.1}$$

due to 1.24. This equation tells us that the norm map $N_{k,\Phi} : T^k \to T^E$ factors through T which is defined as the pullback of the following square:

$$\begin{array}{ccc} T & \longrightarrow & T^E \\ \downarrow & \lrcorner & \downarrow \\ \mathbb{G}_m & \longrightarrow & T^F \end{array}$$

Here F is the largest totally real subalgebra of E and the map $T^E \to T^F$ is given by $a \mapsto a \cdot \iota_E a$. In particular, note that

$$T(R) = \{a \in (E \otimes_{\mathbb{Q}} R)^{\times} \mid a \cdot \iota_E(a) \in R^{\times}\}$$

for any $\mathbb{Q}$-algebra R.

Lemma 5.4 (Milne)**.** *Let $\mathbb{A}_f$ denote the finite adeles of $\mathbb{Q}$ and let $\mathbb{A}_{f,E}^\times$ denote the finite idèles of E. Then the map*

$$T(\mathbb{A}_f)/T(\mathbb{Q}) \to T^E(\mathbb{A}_f)/T^E(\mathbb{Q}) = \mathbb{A}_{f,E}^\times/E^\times$$

defined by $T \to T^E$ is injective and makes the left hand side a topological subgroup of the right hand side.

Proof. Let $a \in T(\mathbb{A}_f)$ which is mapped into $T^E(\mathbb{Q})$ along the map. Note that $T(\mathbb{A}_f) \to T^E(\mathbb{A}_f)$ is just the inclusion map, so a is an element of $\mathbb{A}_{f,E}^\times$. But since it is mapped into $T^E(\mathbb{Q}) = E^\times$, we see that it must have been an element of $E^\times \cap T(\mathbb{A}_f)$, but this is the same as $T(\mathbb{Q})$.

□

Lemma 5.5 (Milne)**.** *The space $T(\mathbb{A}_f)/T(\mathbb{Q})$ is Hausdorff.*

Proof. Lemma 9.13 in [3]. □

5.1.3 Polarisations

Let A be an abelian variety with CM by E over $\bar{\mathbb{Q}}$, let $\Phi \subset \mathrm{Hom}(E, \bar{\mathbb{Q}})$ be the CM-type of A. Recall from 2.7 the definition of a rational Riemann form. Also recall that for any such form, there exists an $k \in \mathbb{Z}$ such that $k\psi$ takes values in $\mathbb{Z}$ on $H_1(A, \mathbb{Z})$. We replace ψ by $k\psi$ and assume that ψ takes integer values on $\Lambda := H_1(A, \mathbb{Z})$ for now. Then we can define a bilinear pairing $\psi_m : \frac{1}{m}\Lambda/\Lambda \times \frac{1}{m}\Lambda/\Lambda \to \mu_m(\mathbb{C})$ via

$$\psi_m(x, y) = \exp(2\pi i \psi(x, y) \cdot m) \tag{5.2}$$

Note that since x, y are in $\frac{1}{m}\Lambda$, we see that $\psi(x, y) \in \frac{1}{m^2}\mathbb{Z}$, therefore $\psi(x, y) \cdot m \in \frac{1}{m}\mathbb{Z}$. Also note that for any $x, y \in \frac{1}{nm}\Lambda$, we have

$$\psi_{nm}(x, y)^m = \exp(2\pi i \psi(x, y) \cdot m^2 n) = \psi_n(mx, my)$$

This allows us to define a map $\widetilde{\psi} : T_f A \times T_f A \to \varprojlim_m \mu_m(\mathbb{C})$ via:

$$\widetilde{\psi}(x, y) := (\psi_m(x_m, y_m))_m \tag{5.3}$$

where $x = (x_m)_m, y = (y_m)_m \in T_f A := \varprojlim A_m = \varprojlim \frac{1}{m}\Lambda/\Lambda$, the inverse limit being over all $m \in \mathbb{Z}$. We can extend this to an $\mathbb{A}_f$-bilinear map

$$\widetilde{\psi} : V_f A \times V_f A \to \mathbb{A}_f(1)$$

via tensoring by $\mathbb{Q}$. Here, $\mathbb{A}_f(1) := \varprojlim \mu_m(\mathbb{C}) \otimes_{\mathbb{Z}} \mathbb{Q}$ We note a couple of facts about $\widetilde{\psi}$ here.

Remark 5.6. *1. The first thing to note is that this extended polarisation agrees with our original polarisation. By this we mean that if we let $i : \Lambda \to T_f(A)$ denote the canonical inclusion given by*

$$x \mapsto \left(\frac{x}{m}\right)_m \in \varprojlim_m A_m$$

and then we look at

$$\widetilde{\psi}(i(x), i(y)) = (\psi_m(x/m, y/m))_m = (e^{2\pi i \psi(x,y)/m})_m \in \varprojlim_m \mu_m(\mathbb{C})$$

If we look at this under the isomorphism to $\hat{\mathbb{Z}}$, we see that the image is $(\psi(x, y) \pmod m)_m \in \hat{\mathbb{Z}}$ which is indeed the image of $\psi(x, y)$ under the canonical inclusion $\mathbb{Z} \hookrightarrow \hat{\mathbb{Z}}$.

2. $\mathbb{A}_f(1)$ *has an action of* $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ *where any element* $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ *acts on an element* $(\zeta_n^{a_n})_n \otimes q$ *via*

$$(\zeta_n^{a_n})_n \otimes q \mapsto (\zeta_n^{a_n \chi(\sigma)})_n \otimes q$$

where $\zeta_n := e^{2\pi i/n}$ *for all* $n \in \mathbb{N}$. *Therefore, for a rational number* $q \in \mathbb{Q}$, *the maps* $\mathbb{Q} \hookrightarrow \mathbb{A}_{f,\mathbb{Q}} \cong \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{A}_f(1)$ *act on* q *as:*

$$q \mapsto (q)_p \mapsto (1)_p \otimes q \mapsto (1)_n \otimes q$$

which means that on $\mathbb{Q}$, *the action of* σ *is trivial.*

3. *For any* $\sigma \in \mathrm{Gal}(\mathbb{Q}^{\mathrm{al}}/\mathbb{Q})$, *we see that there is an isomorphism* $\sigma : V_f(A) \to V_f(\sigma A)$ *and we can define*

$$(\sigma\widetilde{\psi})(\sigma x, \sigma y) := \sigma(\widetilde{\psi}(x, y))$$

Which means that for $x, y \in H_1(A, \mathbb{Q})$, *since we have* $\widetilde{\psi}(x, y) \in \mathbb{Q} \subset \mathbb{A}_{f,\mathbb{Q}}$, *we will get* $(\sigma\widetilde{\psi})(\sigma x, \sigma y) \in \mathbb{Q}$. *So we can define a Riemann pairing*

$$\sigma\psi : H_1(\sigma A, \mathbb{Q}) \times H_1(\sigma A, \mathbb{Q}) \to \mathbb{Q}$$

by $(\sigma\psi)(\sigma x, \sigma y) := \psi(x, y)$.

5.2 Ideal theoretic formulation of the main theorem

We first prove a simpler version of the main theorem that will help us prove the full (idelic) version in the next section.

Let A be an abelian variety with CM by a CM algebra E over $\bar{\mathbb{Q}}$ with CM-type Φ. Assume $\mathrm{End}(A) \cap E = \mathcal{O}_E$ and fix a positive integer m. Then we can define a map $\xi : \mathrm{Gal}(\bar{\mathbb{Q}}/E^*) \to C_m(E)$ as follows. For any $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/E^*)$, σA has the CM-type (E, Φ), so σA is isogenous to A, thus there is an $\mathfrak{a}$-multiplication $\alpha : A \to \sigma A$ (via 3.19) for some ideal $\mathfrak{a} \in \mathcal{O}_E$. We know $\deg(\alpha) = N(\mathfrak{a})$ by 3.17. We can replace α by $\alpha \circ a$ for some $a \in \mathfrak{a}^{-1}$ to get $\deg(\alpha)$ coprime to m. Here we use the lemma:

Lemma 5.7 (Milne)**.** *Let* $\mathfrak{a}$ *be a fractional ideal of* E. *For any positive integer* m, *there exists an* $a \in E^\times$ *such that* $a\mathfrak{a} \subset \mathcal{O}_E$ *and* $N(a\mathfrak{a})$ *is coprime to* m.

Proof. Lemma 9.1 in [3]. □

Thus we can assume that α maps A_m to σA_m isomorphically. We define

$$T_m(A) := \prod_{\ell | m} T_\ell(A)$$

This is a rank 1 free $\mathcal{O}_{E,m}$-module, where $\mathcal{O}_{E,m} := \mathcal{O}_E \otimes_{\mathbb{Z}} \prod_{\ell|m} \mathbb{Z}_\ell$, this is because of the proof of 3.5, where we showed that $T_\ell A$ is a rank one $\mathcal{O}_{E,\ell}$-module when $\ell \nmid [\mathcal{O}_E : R]$. Therefore we have two $\mathcal{O}_{E,m}$-isomorphisms

$$T_m(A) \to T_m(\sigma A)$$

one given by $x \mapsto \sigma x$ and another by $x \mapsto \alpha x$. These differ by a homothety by an element $\beta \in \mathcal{O}_{E,m}^\times$, i.e.,

$$\alpha(\beta x) = \sigma x \ \forall \ x \in T_m A$$

For any $b \in \mathcal{O}_E$ sufficiently close to β, we have

$$\alpha(bx) = \sigma(x) \pmod{m}$$

Replace α by $\alpha \circ b$ to get

$$\alpha(x) = \sigma(x) \pmod{m} \ \forall \ x \in T_m A$$

Note that if we chose a different α originally, it would differ from this one by an element of $E^\times$ according to 3.20. So this α is an $\mathfrak{a}$-multiplication for an ideal $\mathfrak{a}$ well-defined up to an element of $E^\times$ of the form $b = 1 \pmod{m}$. So, from any σ we get a well defined element of $C_m(E)$, the ray class group of E with modulus m. We can show that this map is a group homomorphism. Let $\sigma' \in \mathrm{Gal}(\bar{\mathbb{Q}}/E^*)$ be another element and let $\alpha' : A \to \sigma' A$ be an $\mathfrak{a}'$-multiplication acting as σ' on A_m. Then

$$\sigma\alpha' : \sigma A \to \sigma'\sigma A$$

is an $\mathfrak{a}'$-multiplication and we get $\sigma\alpha' \circ \alpha : A \to \sigma'\sigma A$ is an $\mathfrak{a}\mathfrak{a}'$-multiplication acting as $\sigma'\sigma$ on A_m and so we get

$$\sigma \mapsto \mathfrak{a}(\sigma)$$

defines a group homomorphism $\mathrm{Gal}(\bar{\mathbb{Q}}/E^*) \to C_m(E)$. Since this is a map to an abelian group, it factors through some $\mathrm{Gal}(k/E^*)$ with k/E^* an abelian extension. So we can take $k = E^*(\mathfrak{m})$, that is a ray class field for some modulus $\mathfrak{m}$. So by Class Field Theory we get a map

$$C_{\mathfrak{m}}(E^*) \to C_m(E)$$

We now have the proposition:

Proposition 5.8 (Milne)**.** *Let A be an abelian variety with CM by a CM algebra E over $\bar{\mathbb{Q}}$ with CM-type Φ. Let $\mathrm{End}(A) \cap E = \mathcal{O}_E$ and fix a positive integer m. Then the map we constructed above is given by N_Φ.*

Proof. It suffices to check this on generators of $C_{\mathfrak{m}}(E^*)$. Each generator has infinitely many prime ideals in its equivalence class due to the Chebotarev density theorem (§V.2.5 of [4]). So we can pick representatives of these generators that are primes where A has good reduction, and where E and E^* are unramified. Then using 4.8.1, we see that $\mathfrak{a}(\sigma) = N_\Phi(\mathfrak{p})$ as desired. □

5.3 Idelic formulation of the main theorem

Let A be an abelian variety with CM by E over $\bar{\mathbb{Q}}$, let $\Phi \subset \mathrm{Hom}(E, \bar{\mathbb{Q}})$ be the CM-type of A. For any $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/E^*)$, there exists an E-isogeny $\alpha : A \to \sigma A$, since A and σA have the same CM-type. Here we use 2.24 and 1.13. Now we have two maps $V_f(\alpha), V_f(\sigma) : V_f(A) \to V_f(\sigma A)$ defined by

$$V_f(\sigma)(x) := \sigma x, \quad V_f(\alpha)(x) := \alpha x$$

Since $V_f(A)$ is a rank 1 module over $\mathbb{A}_{f,E}$ (due to 3.5), therefore we see that there exists an element $\eta(\sigma) \in \mathbb{A}_{f,E}^\times$ such that

$$\sigma(x) = \alpha(\eta(\sigma)x)$$

When the choice of α changes, 3.20 tells us $\eta(\sigma)$ changes only by an element of $E^\times$. So we get a well defined map

$$\eta : \mathrm{Gal}(\bar{\mathbb{Q}}/E^*) \to C_E$$

where $C_E := \mathbb{A}_{f,E}^\times / E^\times$ is the idele class group. The main theorem then states:

Theorem 5.9 (Milne)**.** *Let A be an abelian variety with CM by E over $\bar{\mathbb{Q}}$, let $\Phi \subset \mathrm{Hom}(E, \bar{\mathbb{Q}})$ be the CM-type of A. Let $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/E^*)$ then, for any $s \in \mathbb{A}_{f,E^*}^\times$ with $\mathrm{art}_{E^*}(s) = \sigma|_{(E^*)^{ab}}$, there is a unique E-"isogeny" $\alpha : A \to \sigma A$ such that*

$$\alpha(N_\Phi(s) \cdot x) = \sigma(x)$$

for all $x \in V_f(A)$.

Here we put isogeny in quotes, because the map is only a rational multiple of an isogeny, and is not an honest isogeny. We start the proof with a lemma:

Lemma 5.10 (Milne)**.** *Let $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ and $s \in \mathbb{A}_{f,E^*}^\times$ such that $\mathrm{art}_{E^*}(s) = \sigma|_{E^{*ab}}$. Let $\eta(\sigma) \in \mathbb{A}_{f,E}^\times$ such that*

$$\alpha(\eta(\sigma)x) = \sigma(x) \ \forall \ x \in V_f A$$

for some E-"isogeny" $\alpha : A \to \sigma A$. Then

$$\eta(\sigma)/N_\Phi(s) \pmod{E^\times} \in T(\mathbb{A}_f)/T(\mathbb{Q})$$

Proof. We know that

$$N_\Phi(s) \cdot \iota_E N_\Phi(s) = \mathrm{Nm}_{\mathbb{A}_{f,E^*}/\mathbb{A}_f}(s) = \chi(\sigma) \cdot a$$

for some $a \in \mathbb{Q}_{>0}$. Here we used lemma 5.3 in the last equality and equation (5.1) in the second last equality. Now note that $(\sigma\psi)(\sigma x, \sigma y) := \sigma(\psi(x, y)) = \chi(\sigma) \cdot \psi(x, y)$. But $\sigma x = \alpha(\eta(\sigma)x)$ so we get

$$\begin{aligned}(\sigma\psi)(\alpha(\eta(\sigma)x), \alpha(\eta(\sigma)y)) &= (\sigma\psi)(\eta(\sigma)\alpha(x), \eta(\sigma)\alpha(y)) \\ \implies \chi(\sigma) \cdot \psi(x, y) &= (\sigma\psi)(\eta(\sigma)\overline{\eta(\sigma)}\alpha x, \alpha y)\end{aligned}$$

We wish to prove that $\chi(\sigma)/\eta(\sigma)\overline{\eta(\sigma)}$ is an element of $F_{\gg 0}$, i.e., it is a totally positive element of F, the maximal totally real subalgebra of E. To do this, we want to use 2.20. To this end, we note that for $x, y \in H_1(A, \mathbb{Q})$, and for a chosen isomorphism $\theta : E \to H_1(A, \mathbb{Q})$ and corresponding $t \in E^\times$, we have:

$$\begin{aligned}\chi(\sigma)\mathrm{Tr}_{E/\mathbb{Q}}(t\theta^{-1}x \cdot \overline{\theta^{-1}y}) &= (\sigma\psi)(\eta(\sigma)\overline{\eta(\sigma)}\alpha x, \alpha y) \\ \implies (\sigma\psi)(\alpha x, \alpha y) &= \mathrm{Tr}_{E/\mathbb{Q}}\left(t\theta^{-1}\left(\frac{\chi(\sigma)}{\eta(\sigma)\overline{\eta(\sigma)}} \cdot x\right) \cdot \overline{\theta^{-1}y}\right)\end{aligned}$$

Here we think of $\mathrm{Tr}_{E/\mathbb{Q}}$ as an extension of the usual trace map $E \to \mathbb{Q}$ to a map $\mathbb{A}_{f,E} \to \mathbb{A}_{f,\mathbb{Q}}$ simply via tensoring. Similarly, the parametrisation θ is thought of as an $\mathbb{A}_{f,E}$-linear isomorphism $\theta : \mathbb{A}_{f,E} \to V_f(A)$.

We know for $x, y \in H_1(A, \mathbb{Q})$, the left hand side is in $\mathbb{Q}$ by (3) of 5.6. Also, let $t' \in E^\times$ be such that $\sigma\psi(\alpha x, \alpha y) = \mathrm{Tr}_{E/\mathbb{Q}}(t'\theta^{-1}x \cdot \overline{\theta^{-1}y})$. Then we have

$$t'\theta^{-1}(x) = t\theta^{-1}\left(\frac{\chi(\sigma)}{\eta(\sigma)\overline{\eta(\sigma)}} \cdot x\right)$$

since the trace pairing is non degenerate. Since θ is E-linear, we get

$$t \cdot \frac{\chi(\sigma)}{\eta(\sigma)\overline{\eta(\sigma)}} \cdot x = t' \cdot x$$

which implies $\chi(\sigma)/\eta(\sigma)\overline{\eta(\sigma)} = t'/t \in E$. We see by 2.20 that

$$\chi(\sigma)/\eta(\sigma)\overline{\eta(\sigma)} \in F_{\gg 0}$$

Now note that $N_\Phi(s) \in T(\mathbb{A}_f)$ since $N_\Phi(s) \cdot \overline{N_\Phi(s)} \in \mathbb{A}_f$. We want that $\eta(\sigma)$ also be an element of the Hausdorff subgroup $T(\mathbb{A}_f)/T(\mathbb{Q})$ of C_E. For this, we already know that there exists a $c \in F_{\gg 0}$ such that

$$c = \chi(\sigma)/\eta(\sigma)\overline{\eta(\sigma)}$$
$$\implies ac = \gamma\bar{\gamma}$$

where $a = N_\Phi(s)\overline{N_\Phi(s)}/\chi(\sigma) \in \mathbb{Q}_{>0}$ and $\gamma := N_\Phi(s)/\eta(\sigma)$. Being a totally positive element, ac is a local norm of E at all infinite primes. Since it is also equal to $\gamma\bar{\gamma}$, it is a local norm at all finite primes as well. Therefore, it is a global norm by the Hasse norm theorem.

Theorem 5.11. *(Hasse norm theorem) If L/K is a cyclic extension of number fields, then if a nonzero element of K is a local norm everywhere, then it is a global norm.*

Proof. Refer to §VIII.3.1 of [4]. □

Therefore, there is an element $e \in E^\times$ such that

$$\gamma\bar{\gamma} = e\bar{e}$$

Which implies that $\gamma/e \in T(\mathbb{A}_f)$ and is the same as γ inside $\mathbb{A}_{f,E}^\times/E^\times = C_E$. In other words, $\gamma = \gamma/e \pmod{E^\times}$ inside $T(\mathbb{A}_f)/T(\mathbb{Q})$. Also note that $\eta : \mathrm{Gal}(\bar{\mathbb{Q}}/E^*) \to C_E$ factors through $\mathrm{Gal}(\bar{\mathbb{Q}}/E^*)^{\mathrm{ab}}$ and we get

$$\eta' : C_{E^*} \to C_E$$

by Class Field theory. This gives $\gamma/e = N_\Phi(s)/e\eta'(s) = N_\Phi(s)/e\eta(\sigma) \in T(\mathbb{A}_f)$. Which implies that $\gamma = N_\Phi(s)/\eta(\sigma) \pmod{E^\times}$ in $T(\mathbb{A}_f)/T(\mathbb{Q})$ as required. □

Now we are ready to finish the proof of 5.9.

Proof. Following the proof of 5.10, we know that $N_\Phi(s)/\eta(\sigma) \in T(\mathbb{A}_f)/T(\mathbb{Q}) \subset C_E$. We want to show that $\gamma = 1$. This we do by showing that $N_\Phi(s) \in \eta'(s)U$ for all open sets $U \subset C_E$ containing 1. It is enough to show this for a family of open sets $U \subset T(\mathbb{A}_f)/T(\mathbb{Q})$ such that their intersection is $\{1\}$. It is known that

$$\ker(C_E \to \mathrm{Gal}(E^{\mathrm{ab}}/E)) = \bigcap_m U(m) = \overline{\{1\}}$$

where $U(m) := \ker(C_E \to C_m(E))$, for $m \in \mathbb{N}_{>0}$, and $C_m(E)$ denotes the ray class group of E with modulus m. Therefore, it is enough to show that N_Φ and η' agree on the lower arrow in the commutative diagram:

$$\begin{array}{ccc} C_{E^*} & \longrightarrow & C_E \\ \downarrow & & \downarrow \\ C_{\mathfrak{m}}(E^*) & \longrightarrow & C_m(E) \end{array}$$

But this follows from 5.8. □

5.4 Main theorem in terms of Uniformisations

Let (A, i, ψ) be a CM abelian variety with a polarisation ψ. A choice of a basis element $e \in H_1(A, \mathbb{Q})$ determines a uniformisation

$$\theta : \mathbb{C}^\Phi \to A(\mathbb{C})$$

and hence a quadruple $(E, \Phi, \mathfrak{a}, t)$ called the type of (A, i, ψ) relative to θ (see 2.25). We can formulate our last theorem in terms of such uniformisations, and this is what will really be used in constructing our abelian extensions later.

Theorem 5.12 (Milne)**.** *Let (A, i, ψ) be of type $(E, \Phi, \mathfrak{a}, t)$ relative to $\theta : \mathbb{C}^\Phi \to A(\mathbb{C})$. Let $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/E^*)$. For any $s \in \mathbb{A}_{f,E^*}^\times$ such that $\mathrm{art}_{E^*}(s) = \sigma|_{E^{*\mathrm{ab}}}$, there exists a unique $\theta' : \mathbb{C}^\Phi \to \sigma A(\mathbb{C})$ such that $\sigma(A, i, \psi)$ has type $(E, \Phi, f\mathfrak{a}, t\chi(\sigma)/f\bar{f})$ where $f := N_\Phi(s) \in \mathbb{A}_{f,E^*}^\times$.*

Proof. We know by 5.9 that there exists an isogeny $\alpha : A \to \sigma A$ such that

$$\alpha(N_\Phi(s)x) = \sigma x \ \forall \ x \in V_f A$$

Then

$$H_1(\alpha) : H_1(A, \mathbb{Q}) \xrightarrow{\cong} H_1(\sigma A, \mathbb{Q})$$

Let θ' be the uniformisation of σA defined by $H_1(\alpha)(e)$, where e is the image of 1 under $\theta : E \to H_1(A, \mathbb{Q})$. We also know by the proof of 5.10, that if the polarisation ψ on A is given by $t \in E^\times$, i.e.,

$$\psi(ex, ey) = \mathrm{Tr}_{E/\mathbb{Q}}(tx\bar{y})$$

then the polarisation $\sigma\psi$ on σA is given by the element $t' := t\chi(\sigma)/N_\Phi(s)\overline{N_\Phi(s)} = t\chi(\sigma)/f\bar{f}$, relative to θ'. We also have the commutative diagram

$$\begin{array}{ccc} V_f(A) = \mathbb{A}_f \otimes_{\mathbb{Q}} H_1(A, \mathbb{Q}) & \xrightarrow{\ \sigma\ } & V_f(\sigma A) = \mathbb{A}_f \otimes_{\mathbb{Q}} H_1(\sigma A, \mathbb{Q}) \\ {\scriptstyle N_\Phi(s)}\downarrow & \nearrow_{\alpha} & \\ \mathbb{A}_f \otimes_{\mathbb{Q}} H_1(A, \mathbb{Q}) & & \end{array}$$

which restricts to the following commutative diagram since α and σ restrict to $H_1(A, \mathbb{Q})$ and map to $H_1(\sigma A, \mathbb{Q})$ isomorphically, which implies that $N_\Phi(s)$ restricted to $H_1(A, \mathbb{Q})$ must also be an isomorphism to $H_1(A, \mathbb{Q})$.

$$\begin{array}{ccc} H_1(A, \mathbb{Q}) & \xrightarrow{\ \sigma\ } & H_1(\sigma A, \mathbb{Q}) \\ {\scriptstyle N_\Phi(s)}\downarrow & \nearrow_{\alpha} & \\ H_1(A, \mathbb{Q}) & & \end{array}$$

where each arrow is an E-isomorphism. If $e \in H_1(A, \mathbb{Q})$ is the chosen basis element, then $H_1(A, \mathbb{Z}) = \mathfrak{a}e$ and we get $N_\Phi(s)H_1(A, \mathbb{Z}) = N_\Phi(s)\mathfrak{a}e$ as its image under $N_\Phi(s)$. The image of this under α matches the image of $H_1(A, \mathbb{Z})$ under σ, which is $H_1(\sigma A, \mathbb{Z})$. If we pick $\alpha(e) \in H_1(\sigma A, \mathbb{Q})$ as our basis element, then $H_1(\sigma A, \mathbb{Z}) = \alpha(N_\Phi(s)H_1(A, \mathbb{Z})) = \mathfrak{a}N_\Phi(s)\alpha(e)$. This gives us that the type of $\sigma(A, i, \psi)$ is given by $(E, \Phi, f\mathfrak{a}, \chi(\sigma)t/f\bar{f})$ as desired. □

Chapter 6

Construction of Class Fields

Until now we have developed a substantial amount of the theory of CM-types and CM abelian varieties, mostly following Milne. Now we are ready to switch gears and give a construction of the Hilbert Class field and the Ray class fields of the reflex field of a CM-pair, following Shimura's work in [6] .

6.1 Field of Moduli

Let A be an abelian variety over $\mathbb{C}$ and let ψ be a rational Riemann form on A.

Definition 6.1. *For any two pairs (A_1, ψ_1) and (A_2, ψ_2), a morphism (respectively, an isomorphism) of polarised abelian varities is a morphism (respectively, an isomorphism) $\varphi : A_1 \to A_2$ of abelian varieties such that $\varphi^*\psi_2 := \psi_2 \circ \varphi = \psi_1 \pmod{\mathbb{Q}_{>0}}$.*

In particular, the above definition implies that $(A, \psi_1) \cong (A, \psi_2)$ if $\psi_1 = \psi_2 \pmod{\mathbb{Q}_{>0}}$.

We hope to have an analog of the j-invariant for abelian varieties, that will allow us to construct the Hilbert Class field of our CM-field, just as the j-invariant gives the Hilbert Class field in case of an imaginary quadratic field. But in general the existence of such a function is not expected. Instead we have something called a field of moduli, as Shimura defines it in [6].

Proposition 6.2 (Shimura)**.** *Let $(A, \mathcal{C})$ be a polarised abelian variety over k. Then there exists a field $k_0 \subset k$ such that for any $\sigma : k \to k'$ an isomorphism of fields, $(A, \mathcal{C}) \cong (\sigma A, \sigma\mathcal{C})$ iff $\sigma|_{k_0} = \mathrm{id}_{k_0}$. In addition, any field of definition of $(A, \mathcal{C})$ contains k_0.*

Proof. Proposition 5 in [7]. □

It is to be noted that the above proposition has been proved with the notion of polarisations that Shimura uses, and not the one that we have been working with here. So to use Shimura's proof of the existence of field of moduli in 6.15 and 6.20, one must first show that the two notions of polarisation are equivalent and that the action of $\sigma \in \mathrm{Gal}(\mathbb{C}/E^*)$ coincides as well. To this end, let us first recall the notion of polarisation due to Shimura.

Definition 6.3. *Let V be a complete variety defined over a field k and X be an ample divisor on it that is rational over k. Then denote by $\mathcal{C}(X) := \{Y \mid \exists\ m, m' \in \mathbb{N}$ such that $mX \sim m'Y\}$ where $\sim$ denotes algebraic equivalence of divisors. This $\mathcal{C}(X)$ is called a polarisation on V.*

Let $\sigma : k \to k'$ be an isomorphism of fields, then we get a pair $(\sigma V, \sigma\mathcal{C})$ of a polarised variety where $\sigma\mathcal{C} := \mathcal{C}(\sigma X)$. Now replace V by A, an abelian variety and k by $\mathbb{C}$, and now we would like

to show that upto isomorphism, pairs of the form $(A, \mathcal{C})$ are in 1-1 correpondence with pairs of the form (A, ψ).

To see this, note that Riemann forms ψ on $\mathbb{C}^n/\Lambda$ arise from and give rise to holomorphic theta functions f on $\mathbb{C}^n$ with respect to the lattice Λ. Although this correspondence is not 1-1, we do know that the Riemann form depends only on the algebraic equivalence class of the divisor X of any theta function f associated to it. In other words, Riemann forms ψ are in 1-1 correspondence with effective divisors X on A upto algebraic equivalence. Now these have to be non-degenerate since ψ gives an isogeny $A \to A^\vee$, hence X has to give an isogeny $\varphi_X : A \to A^\vee$ as well. Here, $\varphi_X(u) := \mathrm{Cl}(X_u - X)$, see the following paragraphs for why the two isogenies agree. (All this is due to the discussion in §3.1 of [6]).

Therefore, Riemann forms ψ upto a factor in $\mathbb{Q}_{>0}$ are in 1-1 correspondence with non degenerate effective divisors on A upto algebraic equivalence and a factor in $\mathbb{Q}_{>0}$. But by proposition 13 of §4.2 of [6], we know that effective non degenerate divisors are exactly the effective ample divisors. Therefore, the set of $\psi \pmod{\mathbb{Q}_{>0}}$ is in 1-1 correspondence with the set of effective ample divisors on A upto algebraic equivalence and $\mathbb{Q}_{>0}$. But since for any ample divisor, some positive integer multiple of it is linearly equivalent to an effective divisor, we see that the set of effective ample divisors on A upto algebraic equivalence and $\mathbb{Q}_{>0}$ is in 1-1 correspondence with the set of $\mathcal{C}(X)$ as X runs over ample divisors. In the end, we have the set of ψ upto a positive rational factor is in 1-1 correspondence with the set of $\mathcal{C}(X)$ as it runs over ample divisors. Now we just need to show that the action of $\sigma \in \mathrm{Gal}(\mathbb{C}/E^*)$ coincides on both sides.

We know $\varphi_X(u) = \mathrm{Cl}(X_u - X)$ which is a divisor on A that is algebraically equivalent to 0. Let f be a normalized holomorphic theta function on $\mathbb{C}^n/\Lambda$ which is an analytic representation of A, given by an isomorphism $\theta : A \to \mathbb{C}^n/\Lambda$ (refer to 2.24). Choose f such that $(f) = \theta(X)$. We have a Riemann form ψ associated to f. Now define

$$\Phi_u(x) := f(x-u)f(x)^{-1}e^{2\pi i H(x,u)}$$

where $H(x,u) := 1/2(E(x,u) - iE(x,iu))$. Note that $(\Phi_u) = X_u - X$. Then we see that

$$\Phi_u(x+d) = \Phi_u(x)e^{2\pi i\psi(u,d)} \ \forall \ d \in \Lambda$$

So, if we now look at σX on σA, we see that the theta function changes into $\sigma \circ f \circ \sigma^{-1}$. One sees that this is still holomorphic and satisfies the theta function relation with respect to the lattice $\sigma\Lambda$. Indeed, one can check holomorphicity by looking at the Taylor expansion around any point. Then $\varphi_{\sigma Y}(\sigma u) = \mathrm{Cl}(\sigma X_{\sigma u} - \sigma X)$. The associated theta function becomes $\Phi_{\sigma u} = \sigma \circ \Phi_u \circ \sigma^{-1}$. Thus we have

$$\Phi_{\sigma u}(\sigma x + \sigma d) = \Phi_{\sigma u}(\sigma x)e^{2\pi i(\sigma\psi)(\sigma u, \sigma d)} \ \forall \ d \in \Lambda$$

Substituting $\Phi_{\sigma u} = \sigma \circ \Phi_u \circ \sigma^{-1}$,

$$\sigma(\Phi_u(x+d)) = \sigma(\Phi_u(x))e^{2\pi i(\sigma\psi)(\sigma u, \sigma d)}$$

which implies that

$$\begin{aligned}
e^{2\pi i(\sigma\psi)(\sigma u,\sigma d)} &= \sigma(e^{2\pi i\psi(u,d)}) \ \forall \ d \in \Lambda \\
\implies \sigma(e^{2\pi i\psi(x_m,y_m)m}) &= e^{2\pi i(\sigma\psi)(\sigma x_m, \sigma y_m)m} \ \forall \ x_m, y_m \in (1/m)\Lambda/\Lambda \\
\implies \sigma(\psi_m(x_m,y_m)) &= (\sigma\psi)_m(\sigma x_m, \sigma y_m) \\
\implies \widetilde{(\sigma\psi)}(\sigma x, \sigma y) &= \sigma(\widetilde{\psi}(x,y)) \\
\implies \widetilde{(\sigma\psi)}(\sigma x, \sigma y) &= \chi(\sigma)\cdot\widetilde{\psi}(x,y)
\end{aligned}$$

showing that the action of σ is what we expected.

6.2 Kummer Varieties

We collect a few results about quotients of abelian varieties from Shimura in this section.

Proposition 6.4 (Shimura)**.** *Let A be an abelian variety along with a finite subgroup $H \subset \mathrm{Aut}(A)$ of the automorphisms of A. Let A and the elements of H be defined over some field k. Then there exists a projective variety W and a rational map $h : A \to W$ such that*

1. *h is defined everywhere on A.*
2. *$h(x) = h(y)$ iff $x = \gamma \cdot y$ for some $\gamma \in H$.*
3. *If $h' : A \to W'$ is another rational map into a variety W' satisfying $h' = h' \circ \gamma$ for all $\gamma \in H$, then there exists a rational map $\phi : W \to W'$ such that $h' = \phi \circ h$ and ϕ is defined at $h(x)$ whenever h' is defined at x.*

Proof. Proposition 16 in §4.3 of [6]. □

Such a projective variety (W, h) is called a quotient of A by G. In general, abelian varieties have huge automorphism groups. But a polarised abelian variety has only finitely many automorphisms.

Proposition 6.5 (Shimura)**.** *Every polarised abelian variety has only finitely many automorphisms.*

Proof. Proposition 17 of §4.4 in [6]. □

Definition 6.6. *Given a polarised abelian variety (A, ψ) with automorphism group G, a **Kummer variety** is a quotient of A by G.*

Now we are ready to define normalised Kummer varieties, this is what we will use in our construction of ray class fields later.

Proposition 6.7 (Shimura)**.** *Let (A, ψ) be a polarised abelian variety and k_0 its field of moduli. Then there exists a Kummer variety (W, h) of (A, ψ) satisfying*

1. *W is defined over k_0.*
2. *h is defined over every field of definition of (A, ψ) containing k_0.*
3. *Given a field of definition k of (A, ψ), let $\sigma : k \to k'$ an isomorphism of fields fixing k_0 (recall every field of definition contains k_0). For any isomorphism $\eta : (A, \psi) \to (\sigma A, \sigma\psi)$, we have $h = (\sigma h) \circ \eta$.*

Remark 6.8.
1. *Such a pair (W, h) is called a **normalised Kummer variety**.*
2. *Note that here, $\sigma h : \sigma A \to \sigma W$ is the quotient map modified by the action of σ. But also, since $\sigma|_{k_0} = \mathrm{id}_{k_0}$, we get $\sigma W = W$.*
3. *The normalised Kummer variety is uniquely determined by (A, ψ) upto biregular birational mappings defined over k_0.*

6.3 Polarisation Type

Let (A, i, ψ) be a fixed principal CM abelian variety over $\bar{\mathbb{Q}}$ with CM-type (E, Φ). We fix this throughout the following sections. Following the discussion of the previous section, one notes that polarisations are only significant upto a scaling by a positive rational. We will use this fact repeatedly.

Choose a uniformiser $e \in H_1(A, \mathbb{Q})$ and let $\mathfrak{a}$ be the lattice ideal of E such that $e\mathfrak{a} = H_1(A, \mathbb{Z})$. Then (A, i, ψ) is of type $(E, \Phi, \mathfrak{a}, t)$ with respect to e, where $t \in E^\times$ describes the polarisation with respect to e. Notice that if we replace e by ae, for some $a \in E^\times$, we see that (A, i, ψ) becomes of type $(E, \Phi, a^{-1}\mathfrak{a}, ta\bar{a})$. The quantity that is preserved in this base change is $\mathfrak{a}\bar{\mathfrak{a}}t$. We define:

Definition 6.9 (Polarisation Type)**.** *Given a polarised principal CM abelian variety (A, i, ψ), with uniformisation type $(E, \Phi, \mathfrak{a}, t)$. We then define the $\mathcal{P}$-type of (A, i, ψ) as*

$$\mathcal{P}(A, i, \psi) := \mathfrak{a}\bar{\mathfrak{a}}t \pmod{\mathbb{Q}_{>0}}.$$

The $\mathcal{P}$-type is a well defined quantity, since changing the uniformisation by the transformation $(E, \Phi, \mathfrak{a}, t) \mapsto (E, \Phi, \mathfrak{a}a, t/a\bar{a})$ leaves the quantity $\mathcal{P}$ invariant.

Lemma 6.10. *Let (A, i, ψ) be a polarised principal CM abelian variety of CM-type (E, Φ), then $\sigma(A, i, \psi)$ has the same $\mathcal{P}$-type as (A, i, ψ), for any $\sigma \in \mathrm{Gal}(\bar{\mathbb{Q}}/E^*)$.*

Proof. Suppose (A, i, ψ) is of type $(E, \Phi, \mathfrak{a}, t)$ relative to some uniformisation $\theta : E \to H_1(A, \mathbb{Q})$. Then $\sigma(A, i, \psi)$ has type $(E, \Phi, \mathfrak{a}N_\Phi(s), \chi(\sigma)t/\mathrm{Nm}_{E^*/\mathbb{Q}}(s))$ where $s \in \mathbb{A}_{f,E^*}^\times$ is such that $\mathrm{art}_{E^*}(s) = \sigma|_{E^{*\mathrm{ab}}}$ by 5.9. We have

$$\mathcal{P}(\sigma(A, i, \psi)) = N_\Phi(s)\mathfrak{a}\overline{N_\Phi(s)\mathfrak{a}}\frac{t\chi(\sigma)}{\mathrm{Nm}_{E^*/\mathbb{Q}}(s)} = \mathfrak{a}\bar{\mathfrak{a}}t\chi(\sigma)$$

But $\chi(\sigma) \in \mathbb{A}_{f,\mathbb{Q}}^\times$, and so it corresponds to an ideal in $\mathbb{Q}$. But since $h(\mathbb{Q}) = 1$, it corresponds to an element of $\mathbb{Q}_{>0}$. Therefore,

$$\mathcal{P}(\sigma(A, i, \psi)) = \mathfrak{a}\bar{\mathfrak{a}}t \pmod{\mathbb{Q}_{>0}}$$

□

6.4 Constructing the Hilbert Class field

The following strategy of constructing the Hilbert Class field has been adapted from [8], where Silverman exposits the construction in the case of CM elliptic curves. We use a similar strategy, generalising to the case of CM abelian varieties by using Shimura's development of CM theory and putting everything in Milne's language.

For this section and the following ones, we fix a field $k \supset E^*$ contained in $\mathbb{C}$. Here, we consider polarisations upto multiplication by a positive rational number, i.e., we identify two tuples (A, i, ψ_1) and (A, i, ψ_2) if $\psi_1 \in \psi_2\mathbb{Q}_{>0}$. Also, $(A_1, i_1, \psi_1) \cong (A_2, i_2, \psi_2)$ iff there is an isomorphism $\varphi : (A_1, i_1) \to (A_2, i_2)$ such that $\varphi^*\psi_2 \in \psi_1 \cdot \mathbb{Q}_{>0}$.

Definition 6.11. *Denote by $I(E, \Phi, \mathcal{P})$ the set of $\bar{k}$-isomorphism classes of polarised principal CM abelian varieties over k of type (E, Φ) along with a polarisation ψ of type $\mathcal{P}$.*

We define the set

$$\mathcal{C}(E) := \{(\mathfrak{a}, \rho) \mid \mathfrak{a} \subset E \text{ is a fractional ideal and } \rho \in E \text{ s.t. } \mathfrak{a}\bar{\mathfrak{a}} = (\rho)\}/\cong$$

where $(\mathfrak{a}_1, \rho_1) \sim (\mathfrak{a}_2, \rho_2)$ iff there exists an element $\mu \in E^\times$ such that $\mathfrak{a}_2 = \mu\mathfrak{a}_1$ and $\rho_2 = \rho_1\mu\bar{\mu}$. Note that $\mathcal{C}(E)$ is a group with identity element $(\mathcal{O}_E, 1)$ and $(\mathfrak{a}, \rho)^{-1} = (\mathfrak{a}^{-1}, \rho^{-1})$.

Lemma 6.12. *The group $\mathcal{C}(E)$ acts on $I(E,\Phi,\mathcal{P})$ via*

$$(E,\Phi,\mathfrak{b},t) \xrightarrow{(\mathfrak{a},\rho)} (E,\Phi,\mathfrak{a}^{-1}\mathfrak{b},t\rho)$$

and this action is free and transitive.

Proof. Note that the $\mathcal{P}$-type of $(E,\Phi,\mathfrak{a}^{-1}\mathfrak{b},t\rho)$ is equal to

$$\mathcal{P}(E,\Phi,\mathfrak{a}^{-1}\mathfrak{b},t\rho) = \mathfrak{a}^{-1}\mathfrak{b}\overline{\mathfrak{a}^{-1}\mathfrak{b}}t\rho = \mathfrak{b}\bar{\mathfrak{b}}t = \mathcal{P}(E,\Phi,\mathfrak{b},t)$$

So the action is well defined.

To show transitivity, assume $(A_1,i_1,\psi_1),(A_2,i_2,\psi_2) \in I(E,\Phi,\mathcal{P})$. Let the two tuples have types $(E,\Phi,\mathfrak{b}_1,t_1)$ and $(E,\Phi,\mathfrak{b}_2,t_2)$ respectively. Then since both of these have the same $\mathcal{P}$-type, we see that

$$\mathfrak{b}_1\overline{\mathfrak{b}_1}t_1 = \mathfrak{b}_2\overline{\mathfrak{b}_2}t_2 \pmod{\mathbb{Q}_{>0}}$$

Therefore, $(\mathfrak{b}_1\mathfrak{b}_2^{-1})\overline{(\mathfrak{b}_1\mathfrak{b}_2^{-1})} = t_2r/t_1$, for some $r \in \mathbb{Q}_{>0}$. So we see that

$$(E,\Phi,\mathfrak{b}_1,t_1) \xrightarrow{(\mathfrak{b}_1\mathfrak{b}_2^{-1},t_2r/t_1)} (E,\Phi,\mathfrak{b}_2,rt_2) \cong (E,\Phi,\mathfrak{b}_2,t_2)$$

as desired.

To show freeness of the action, assume $(E,\Phi,\mathfrak{b},t) \cong (E,\Phi,\mathfrak{a}^{-1}\mathfrak{b},t\rho)$ for some $(\mathfrak{a},\rho) \in \mathcal{C}(E)$. Then there exists an $a \in E^\times$ such that

$$a\mathfrak{b} = \mathfrak{a}^{-1}\mathfrak{b} \text{ and } t/a\bar{a} = t\rho \pmod{\mathbb{Q}_{>0}}$$

Since $\mathfrak{a},\mathfrak{b}$ are ideals of $\mathcal{O}_E$, they are invertible. So we have that $(a) = \mathfrak{a}^{-1}$ and $(\rho) = (a\bar{a})^{-1}$. This implies that $\rho = a^{-1}\bar{a}^{-1} \pmod{\mathcal{O}_E^\times}$. But we already have $\rho = a^{-1}\bar{a}^{-1} \pmod{\mathbb{Q}_{>0}}$, which means that $\rho = a^{-1}\bar{a}^{-1}s$ where $s \in \mathcal{O}_E^\times \cap \mathbb{Q}_{>0} = \{1\}$, therefore, $s = 1$. So we have $\mathfrak{a} = (a^{-1})$ and $\rho = a^{-1}\bar{a}^{-1}$, therefore $(\mathfrak{a},\rho) \cong (\mathcal{O}_E,1)$. □

Using this free and transitive action, one can define a group homomorphism from $\mathrm{Gal}(E^{*\mathrm{ab}}/E^*)$ to $\mathcal{C}(E)$.

Definition 6.13. *We define the map*

$$F : \mathrm{Gal}(E^{*\mathrm{ab}}/E^*) \to \mathcal{C}(E)$$

via $F(\sigma)$ being the unique element of $\mathcal{C}(E)$ such that $\sigma(A,i,\psi) = F(\sigma) * (A,i,\psi)$.

Lemma 6.14. *The map defined in 6.13 is a group homomorphism*

Proof. Transitivity and freeness say that the map is well defined. Now,

$$\begin{aligned} F(\sigma_1\sigma_2) * (A,i,\psi) &= (\sigma_1\sigma_2)(A,i,\psi) \\ &= \sigma_1(\sigma_2(A,i,\psi)) \\ &= F(\sigma_1) * (\sigma_2(A,i,\psi)) \\ &= F(\sigma_1) * (F(\sigma_2) * (A,i,\psi)) \\ &= (F(\sigma_1)F(\sigma_2)) * (A,i,\psi) \end{aligned}$$

Now freeness gives $F(\sigma_1\sigma_2) = F(\sigma_1)F(\sigma_2)$. □

We are now ready to construct the Hilbert class field of E^*.

Theorem 6.15 (Shimura)**.** *Let (A, i, ψ) be a principal CM abelian variety with CM by E over a field $k \supset E^*$ contained in $\mathbb{C}$. Denote by k_0 the field of moduli of (A, i, ψ), then the field extension $k_0^* = k_0E^*/E^*$ is a subfield of the Hilbert Class field H of E^* corresponding to the subgroup*

$$H_0 := \{\mathfrak{a} \in \mathrm{Cl}(E^*) = \mathrm{Gal}(H/E^*) \mid \exists\ \mu \in E \text{ such that } N_\Phi(\mathfrak{a}) = (\mu) \text{ and } \mu\bar{\mu} = N(\mathfrak{a})\}$$

Proof. Recall the map F from 6.13, this has kernel:

$$\begin{aligned}\ker(F) &= \{\sigma \in \mathrm{Gal}(E^{*\mathrm{ab}}/E^*) \mid F(\sigma) = (\mathcal{O}_E, 1)\} \\ &= \{\sigma \mid \sigma(A, i, \psi) \cong (A, i, \psi)\} \\ &= \{\sigma \mid \sigma|_{k_0} = \mathrm{id}_{k_0}\} \\ &= \mathrm{Gal}(E^{*\mathrm{ab}}/k_0E^*)\end{aligned}$$

So, we get an injective map which we will also call F, $F : \mathrm{Gal}(k_0^*/E^*) \hookrightarrow \mathcal{C}(E)$. This implies that k_0^*/E^* is an abelian extension. We want to show that the conductor of k_0^*/E^* is 1. Assume $\mathfrak{m}$ is the said conductor. Then we have by Class field theory,

$$I(\mathfrak{m}) \xrightarrow{\psi} \mathrm{Gal}(k_0^*/E^*) \xrightarrow{F} \mathcal{C}(E)$$

where, $I(\mathfrak{m})$ is the set of fractional ideals in E^* coprime to $\mathfrak{m}$, ψ is surjective and is given by $\mathfrak{p} \mapsto (\mathfrak{p}, k_0^*/E^*)$, where $\mathfrak{p}$ is a prime ideal in $I(\mathfrak{m})$ and $(\mathfrak{p}, k_0^*/E^*)$ is the Frobenius associated to it.

We know by 5.12 and 4.8.1, that $F(\mathfrak{p}, k_0^*/E^*) = (N_\Phi(s)^{-1}, \mathrm{Nm}_{E^*/\mathbb{Q}}(s)^{-1}) = (N_\Phi(\mathfrak{p}), \mathrm{Nm}_{E^*/\mathbb{Q}}(\mathfrak{p}))$, where $s \in \mathbb{A}_{f,E^*}^\times$ such that $\mathrm{art}(s)|_{k_0^*} = (\mathfrak{p}, k_0^*/E^*)$. Therefore, $\mathfrak{p} \in \ker(\psi) \iff N_\Phi(\mathfrak{p}) = (a)$ and $\mathrm{Nm}_{E^*/\mathbb{Q}}(\mathfrak{p}) = a\bar{a}$ for some $a \in E^\times$.

For a principal prime ideal $\mathfrak{p} = (a)$, we have $N_\Phi(\mathfrak{p}) = (N_\Phi(a))$ and we get the element

$$((N_\Phi(a)), \mathrm{Nm}_{E^*/\mathbb{Q}}(a)) \cong (\mathcal{O}_E, 1)$$

Therefore, all principal prime ideals of $I(\mathfrak{m}) \subset \ker(\psi)$. So, all principal prime ideals of $I(\mathfrak{m})$ split completely in k_0^*, meaning that

$$\mathrm{Spl}(k_0^*/E^*) \supset \mathrm{Spl}(H/E^*) \text{ upto finitely many primes}$$

where $\mathrm{Spl}(L/K)$ denotes the set of primes of a number field K that split completely in the finite extension L. Here we use the fact from Class Field Theory, that a prime splits in the Hilbert Class Field iff it is principal.

So we get that $k_0^* \subset H$. Hence the conductor $\mathfrak{m} = (1)$, and we get the map

$$I((1)) \xrightarrow{\psi} \mathrm{Gal}(k_0^*/E^*) \xrightarrow{F} \mathcal{C}(E)$$

with $P((1)) \subset \ker(\psi)$, where $P((1))$ denotes all principal fractional ideals of E^*. So ψ descends to a surjective map on the class group $\mathrm{Cl}(E^*)$ and we get

$$\begin{aligned}\ker\psi &= \{\mathfrak{a} \in \mathrm{Cl}(E^*) \mid (N_\Phi(\mathfrak{a}), \mathrm{Nm}_{E^*/\mathbb{Q}}(\mathfrak{a})) \cong (\mathcal{O}_E, 1)\} \\ &= \{\mathfrak{a} \in \mathrm{Cl}(E^*) \mid \exists\ \mu \in E \text{ such that } N_\Phi(\mathfrak{a}) = (\mu) \text{ and } \mu\bar{\mu} = \mathrm{Nm}_{E^*/\mathbb{Q}}(\mathfrak{a})\} \\ &= \{\mathfrak{a} \in \mathrm{Cl}(E^*) \mid \exists\ \mu \in E \text{ such that } N_\Phi(\mathfrak{a}) = (\mu) \text{ and } \mu\bar{\mu} = N(\mathfrak{a})\} = H_0\end{aligned}$$

thus showing,

$$\psi : \mathrm{Cl}(E^*)/H_0 \xrightarrow{\cong} \mathrm{Gal}(k_0^*/E^*)$$

as desired. □

6.5 Construction of ray class fields of the reflex field

To construct the ray class fields, we use a similar strategy as before, but now we need to also keep track of certain torsion information along with polarisations. To this end, we fix an ideal $\mathfrak{c} \subset \mathcal{O}_E$ and we define:

Definition 6.16.

$$I(E, \Phi, \mathcal{P}, \mathfrak{c}) := \{(A, i, \psi, T) \mid (A, i, \psi) \in I(E, \Phi, \mathcal{P}) \ \& \ T \in \mathrm{Gen}(A[\mathfrak{c}])\}/\cong$$

where $A[\mathfrak{c}] := \ker(\lambda^{\mathfrak{c}} : A \to A^{\mathfrak{c}})$, *and* $\mathrm{Gen}(A[\mathfrak{c}])$ *denotes the set of its generators as an* $\mathcal{O}_E$*-module. The congruence of two elements* (A_1, i_1, ψ_1, T_1) *and* (A_2, i_2, ψ_2, T_2) *is given by an isomorphism* $\varphi : (A_1, i_1, \psi_1) \to (A_2, i_2, \psi_2)$ *such that* $\varphi(T_1) = T_2$.

We also define:

Definition 6.17.

$$\mathcal{C}(E, \mathfrak{c}) := \{(\mathfrak{a}, \rho) \mid \mathfrak{a} \subset E \text{ fractional ideal coprime to } \mathfrak{c} \text{ and } \mathfrak{a}\bar{\mathfrak{a}} = (\rho)\}/\cong$$

where $(\mathfrak{a}_1, \rho_1) \cong (\mathfrak{a}_2, \rho_2)$ *iff* $\exists\ \mu \in E$ *such that* $\mu\mathfrak{a}_1 = \mathfrak{a}_2$ *and* $\mu\bar{\mu}\rho_1 = \rho_2$ *with* $\mu \in P(\mathfrak{c}) :=$ *group of principal fractional ideals* (α) *such that* $\alpha = 1 \pmod{\mathfrak{c}}$.

Now, one sees that $\mathcal{C}(E, \mathfrak{c})$ is an abelian group which acts on $I(E, \Phi, \mathcal{P}, \mathfrak{c})$.

Lemma 6.18. *The group* $\mathcal{C}(E, \mathfrak{c})$ *acts on* $I(E, \Phi, \mathcal{P}, \mathfrak{c})$ *via*

$$(E, \Phi, \mathfrak{b}, t, T) \xrightarrow{(\mathfrak{a}, \rho)} (E, \Phi, \mathfrak{a}^{-1}\mathfrak{b}, t\rho, \lambda^{\mathfrak{a}}T)$$

and this action is free and transitive.

Proof. Note that by 6.12, the action is well defined.

To show transitivity, take two elements (A_1, i_1, ψ_1, T_1) and (A_2, i_2, ψ_2, T_2) of types $(E, \Phi, \mathfrak{b}_1, t_1, T_1)$ and $(E, \Phi, \mathfrak{b}_2, t_2, T_2)$. By the proof of 6.12, we see that $\mathfrak{a} = {\mathfrak{b}_2}^{-1}\mathfrak{b}_1$ and $\rho = t_2/t_1$ work. But now we need to take care of T_1 mapping to T_2.

For this, notice that $(\mathfrak{a}, \rho)$ can be changed into $(\mathfrak{a}\mu, \rho\mu\bar{\mu})$ without affecting the action on $(E, \Phi, \mathfrak{b}_1, t_1)$, but changing T_1's image at the same time. So we choose an element $\mu \in \mathcal{O}_E$ such that $\mu T_1 = T_2$, this is possible since T_1 is a generator of $A[\mathfrak{c}]$. Then $(\mu\mathfrak{a}, \mu\bar{\mu}\rho)$ works.

To show that the action is free, assume that

$$(E, \Phi, \mathfrak{b}, t, T) \xrightarrow{(\mathfrak{a}, \rho)} (E, \Phi, \mathfrak{b}\mathfrak{a}^{-1}, t\rho, \lambda^{\mathfrak{a}}(T)) \cong (E, \Phi, \mathfrak{b}, t, T)$$

This happens iff $\exists\ a \in E^{\times}$ such that

$$\mathfrak{b}a = \mathfrak{b}\mathfrak{a}^{-1} \text{ and } t/a\bar{a} = \rho t \pmod{\mathbb{Q}_{>0}} \text{ and } \lambda^{\mathfrak{a}}T = T$$

This is equivalent to (again by the proof of 6.12),

$$(\mathfrak{a}, \rho) = (a^{-1}, (a\bar{a})^{-1}) \text{ and } \lambda^{\mathfrak{a}}T = T$$

Note that since $\mathfrak{a} = (a^{-1})$, we have $\lambda^{\mathfrak{a}}T = a^{-1}T = T$. But since T is a generator of $A[\mathfrak{c}]$ as an $\mathcal{O}_E/\mathfrak{c}$-module, we see that $a = 1 \pmod{\mathfrak{c}}$. Therefore, $(\mathfrak{a}, \rho) \cong (\mathcal{O}_E, 1)$ in $\mathcal{C}(E, \mathfrak{c})$. □

Now there is a map,

Definition 6.19. *We define the map,*

$$F : \mathrm{Gal}(E^{*\mathrm{ab}}/E^*) \to \mathcal{C}(E, \mathfrak{c})$$

via $F(\sigma)$ being the unique element of $\mathcal{C}(E, \mathfrak{c})$ such that

$$\sigma(A, i, \psi, T) = F(\sigma) * (A, i, \psi, T)$$

One sees in the same way as 6.14 that this is a group homomorphism. Since $\mathcal{C}(E, \mathfrak{c})$ is an abelian group, F passes through a quotient

$$F : \mathrm{Gal}(L/E^*) \hookrightarrow \mathcal{C}(E, \mathfrak{c})$$

for some abelian extension L/E^*. We claim that this is the extension we are looking for.

Theorem 6.20 (Shimura)**.** *Let (E^*, Φ^*) be a primitive CM-pair with (E, Φ) its reflex pair. Let (A, i, ψ) be a polarised principal CM abelian variety with CM by E over a field $k \supset E^*$ contained in $\mathbb{C}$. Denote by k_0 the field of moduli of (A, i, ψ), and $h : A \to W$ be the Weber function that is the quotient map from A to its Kummer variety W. Let b be the smallest positive integer in $\mathfrak{c}$. Let T be a generator of $A[\mathfrak{c}]$ as an $\mathcal{O}_E$-module. Then the field extension $k_0^*(h(T))$ is a subextension of the ray class field $E^*((b))$ associated to (b), corresponding to the subgroup*

$$H_1 := \{\mathfrak{a} \in \mathrm{Cl}(E^*, (b)) \mid \exists\ \mu \in \mathcal{O}_E \text{ such that } N_\Phi(\mathfrak{a}) = (\mu), \mu\bar{\mu} = N(\mathfrak{a}), \mu = 1 \pmod{\mathfrak{c}}\} \subset \mathrm{Cl}(E^*, (b))$$

Note that for this theorem, we need that (E^*, Φ^*) is primitive, in contrast to 6.15.

Proof. We calculate the kernel of F

$$\begin{aligned}
\ker F &= \{\sigma \in \mathrm{Gal}(E^{*\mathrm{ab}}/E^*) \mid \sigma(A, i, \psi, T) \cong (A, i, \psi, T)\} \\
&= \{\sigma \mid \exists\ \eta : \sigma(A, i, \psi) \xrightarrow{\cong} (A, i, \psi)\ \&\ \eta(T) = \sigma(T)\} \\
&= \{\sigma \mid \eta(T) = \sigma(T)\ \&\ \sigma|_{k_0^*} = \mathrm{id}_{k_0^*}\}
\end{aligned}$$

Now note that for any $\sigma \in \ker F$,

$$\sigma(h(T)) = \sigma(h)(\sigma(T)) = \sigma(h)(\eta(T)) = h(T)$$

where the last equality holds due to item 3 of 6.7. This implies that $(E^{*\mathrm{ab}})^{\ker F} \supset k_0^*(h(T))$.

Now, if σ fixes $h(T)$, then

$$\begin{aligned}
&\sigma(h(T)) = h(T) \\
\implies\ &(\sigma h)(\sigma(T)) = (\sigma h)(\eta(T)) \\
\implies\ &\exists\ \pi \in \mathrm{End}(A, i, \psi) \text{ such that } \sigma(T) = \pi(\eta(T))
\end{aligned}$$

So we can replace η by $\pi\eta$ and get $\sigma(T) = \eta(T)$, where η still is an isomorphism $\eta : (A, i, \psi) \to \sigma(A, i, \psi)$. This implies that $\sigma \in \ker F$ and we get

$$(E^{*\mathrm{ab}})^{\ker F} = k_0^*(h(T))$$

So, $L = k_0^*(h(T))$. Now we will try to say something about its conductor.

Let the conductor of L be $\mathfrak{d}$, an ideal of $\mathcal{O}_{E^*}$. Let $\mathfrak{p}$ be any prime ideal in $I(\mathfrak{d})$ such that it satisfies the hypothesis for 4.8.1, then by 5.12 and 4.8.1, we get that $F(\psi(\mathfrak{p})) = (N_\Phi(\mathfrak{p}), N(\mathfrak{p}))$, where $\psi : I(\mathfrak{d}) \to \mathrm{Gal}(L/E^*)$ is the surjective map as given by taking the Frobenius

$$I(\mathfrak{d}) \xrightarrow{\psi} \mathrm{Gal}(L/E^*) \xrightarrow{F} \mathcal{C}(E, \mathfrak{c}).$$

Let b be the smallest positive integer such that $(b) \subset \mathfrak{c}$. Then, for every principal prime ideal (a) in $P((b)) \cap I(\mathfrak{d})$, we have $F(\psi((a)) = ((N_\Phi(a)), N((a)))$. We have that

$$(N_\Phi(a)) = \left(\omega \cdot \prod_{\psi_i \in \Phi^*} \psi_i(a)\right) = \left(\prod_{\psi_i \in \Phi^*} \psi_i(a)\right)$$

where ω is a root of unity, hence a unit in $\mathcal{O}_E$. Since $a = 1 \pmod b$, we have $\psi_i(a) = 1 \pmod b$ for all $\psi_i \in \Phi^*$, and we get $\prod_{\psi_i \in \Phi^*} \psi_i(a) = 1 \pmod b$. Here we used 1.25.2 and this is the only place where we need primitivity of (E^*, Φ^*). This gives us that

$$(N_\Phi(a), N((a))) \cong (\mathcal{O}_E, 1)$$

Therefore, $P((b)) \cap I(\mathfrak{d}) \subset \ker \psi$, so we have

$$\mathrm{Spl}(E^*[(b)]/E^*) \subset \mathrm{Spl}(L/E^*)$$

upto finitely many primes. By Class Field theory, we get $L \subset E^*((b))$. So we have the map now,

$$I((b)) \xrightarrow{\psi} \mathrm{Gal}(L/E^*) \xrightarrow{F} \mathcal{C}(E, \mathfrak{c})$$

where $P((b)) \subset \ker \psi$, so ψ descends to $\mathrm{Cl}(E^*, (b)) \to \mathrm{Gal}(L/E^*)$. The kernel of ψ becomes

$$\begin{aligned} \ker \psi &= \{\mathfrak{a} \in \mathrm{Cl}(E^*, (b)) \mid (N_\Phi(\mathfrak{a}), N(\mathfrak{a})) \cong (\mathcal{O}_E, 1)\} \\ &= \{\mathfrak{a} \in \mathrm{Cl}(E^*, (b)) \mid \exists\ \mu \in \mathcal{O}_E \text{ such that } N_\Phi(\mathfrak{a}) = (\mu), \mu\bar{\mu} = N(\mathfrak{a}), \mu = 1 \pmod{\mathfrak{c}}\} \end{aligned}$$

□

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